Microsoft® Windows

Software Development Kit

Application Style Guide

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Microsoft Corporation

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Introduction

The Microsoft® Windows operating environment is designed to be an interface that makes it easier for the nonprogramming user to use a computer. To fulfill this design, Windows applications should provide a friendly user interface that is consistent with other Windows applications so the user does not have to relearn basic concepts.

This style guide provides suggestions on how to design a consistent user interface for Windows applications. These are reasonable guidelines, intended to encourage a common user interface for all Windows applications, to benefit the user. Applications that employ these guidelines are easier to learn and use.

This guide does not cover all possible user-interface issues. Applications that have needs not covered by this guide should develop logical extensions of these guidelines. Applications should deviate from the guidelines only when absolutely necessary.

The Microsoft Windows operating environment runs with the MS-DOS® operating system. MS-DOS may also be known as DOS or PC-DOS. In this style guide, MS-DOS will be referred to as DOS.

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Main Window

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1.1 Introduction

Every application developed for the Microsoft Windows operating environment should have at least one window to serve as the main visual interface for all user interaction with the application. This main window should be a tiled (nonoverlapping) window with a title bar, System menu, and size box. A menu bar is recommended if the application has commands; scroll bars are recommended if it requires scrolling. The following screen shows the recommended format of the main window:

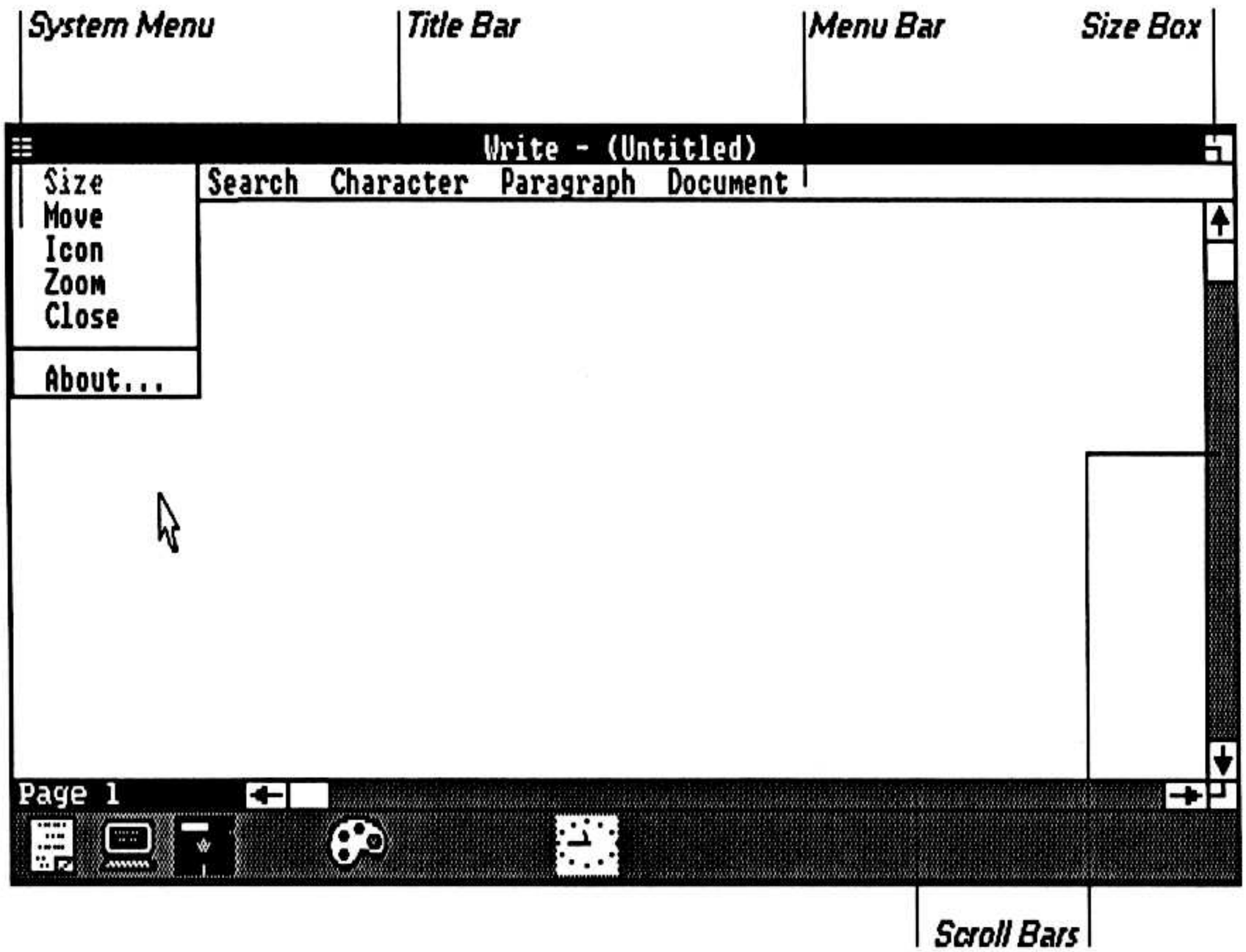


Figure 1.1 Main Window

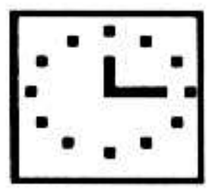
The application should create and display the main window as its first task to let the user know the application is running. It should destroy the main window as its last task.

1.2 Icons

Every application should have an icon that uniquely identifies the application. Windows uses the icon to symbolize the loaded application when that application's main window is not visible.

An icon can be designed in advance and added to an application's executable file as a resource. The application registers the icon with Windows when the main window is created.

An application can also choose to draw the icon while the application is running. This permits an application to continue to display some output even though its main window is not visible. This type of output is limited to the space occupied by the icon. The application must be prepared to update the icon when required. An example is the clock, which continues to display the correct time when it is reduced to an icon, shown as follows:



1.3 Window Captions

To identify an application's window, every application should place a unique caption in the title bar of its main window. At a minimum, the caption should contain the application's name, with the first letter capitalized. Additional text may be added.

For applications whose window content is always associated with a specific data file, the application should add the name of the data file to the caption in the title bar, separating the application name from the filename with a hyphen. The filename should appear in uppercase letters and should include the filename extension (if any). The file's pathname should not be included.

The following screen shows the recommended format for the title-bar caption of the main window:

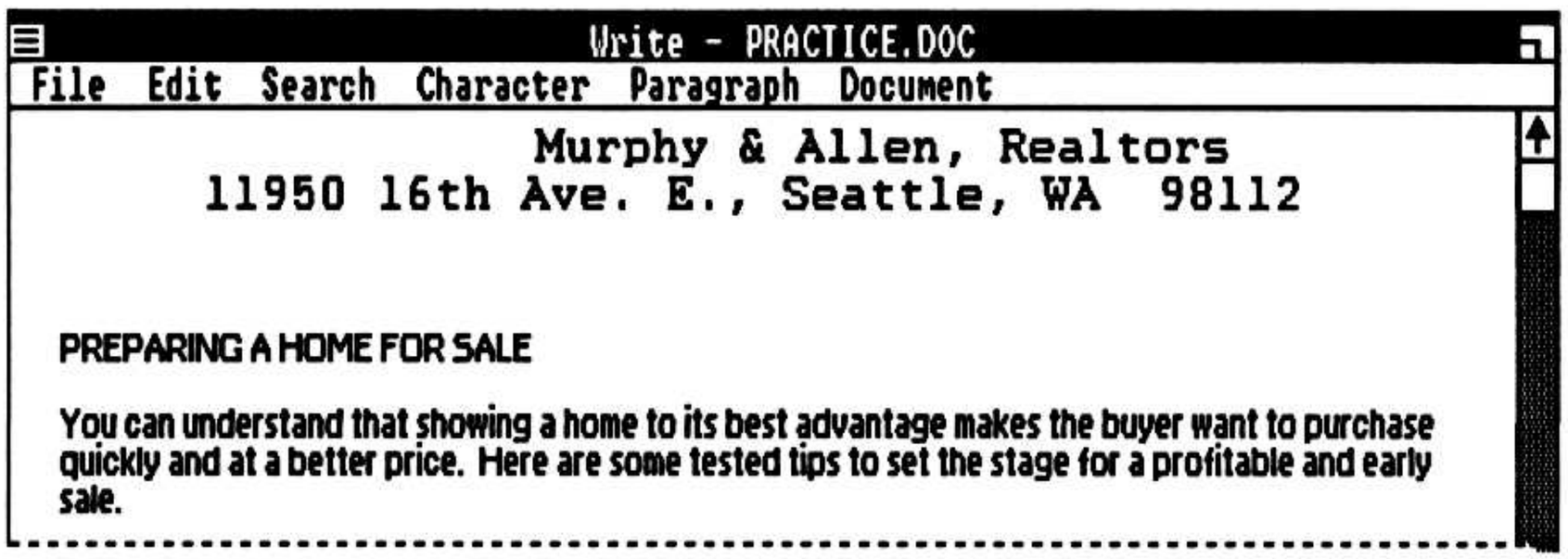


Figure 1.2 Window Caption with Filename

If there is no current file, the string "(Untitled)" should be used, as shown in the following example:

```
Paint - (Untitled)
```

1.4 Output

Applications should display all output, except error messages, warnings, and prompts for additional command information, in the client area of the application's windows. The client area is the rectangle bounded by the menu bar on the top and the scroll bars on the right side and bottom. The client area is the area the user has access to. Output can be text, graphics, or both. The application must decide the format and draw the actual display.

Since client-area size depends on the number of windows on the screen, applications that need to display a complete image (for example, the complete face of a clock) should adjust the image to fit in the available space.

Output too large for the client area is clipped; that is, any part of the output that extends beyond the client area is not displayed. Applications that clip output should provide a method of scrolling the output in the window (see Section 1.5, "Scrolling"). An application must not display any of its output in another application's window.

Applications must be prepared to update the display when requested by Windows.

The following screen shows two applications with output in their windows:

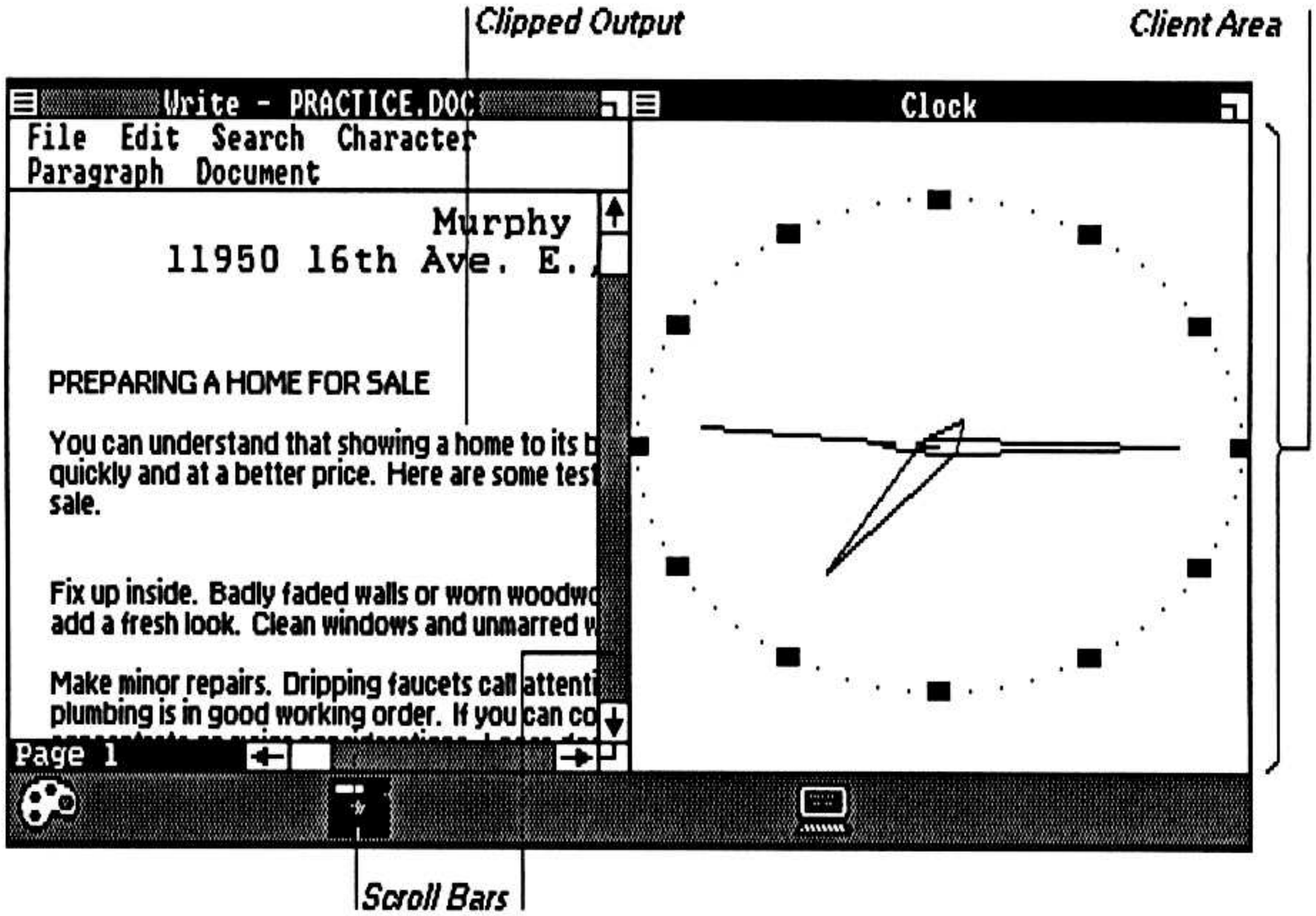


Figure 1.3 Output in Windows

1.5 Scrolling

Applications should provide scrolling for a window's client area whenever the output displayed in the area is too large to display all at once. For example, a window whose output is the contents of a data file can display a portion of the file at a time and let the user scroll the contents of the screen as desired to see other parts of the file.

Applications that scroll should provide ways for scrolling with both the keyboard and the mouse (or other pointing device).

For scrolling with the keyboard, applications should use the following recommended keystrokes:

Keystroke	Action
PAGE UP	Moves contents of window to previous screen.
PAGE DOWN	Moves contents of window to next screen.

For scrolling with a mouse, the application should include a horizontal or vertical scroll bar with the window (it should use both if it permits scrolling in all four directions). Arrows at either end of the scroll bar indicate the direction of scrolling. The application should then respond to user input as follows:

Mouse Movement	Action
Click the Up Arrow	Moves contents of window down one unit.
Click the Down Arrow	Moves contents of window up one unit.
Click the Left Arrow	Moves contents of window right one unit.
Click the Right Arrow	Moves contents of window left one unit.
Click the Scroll Bar	Moves contents of window one screenful.
Drag the Scroll Box	Scrolls to approximate location on bar.

The following screen shows the placement of scroll bars, scroll boxes, and direction arrows:

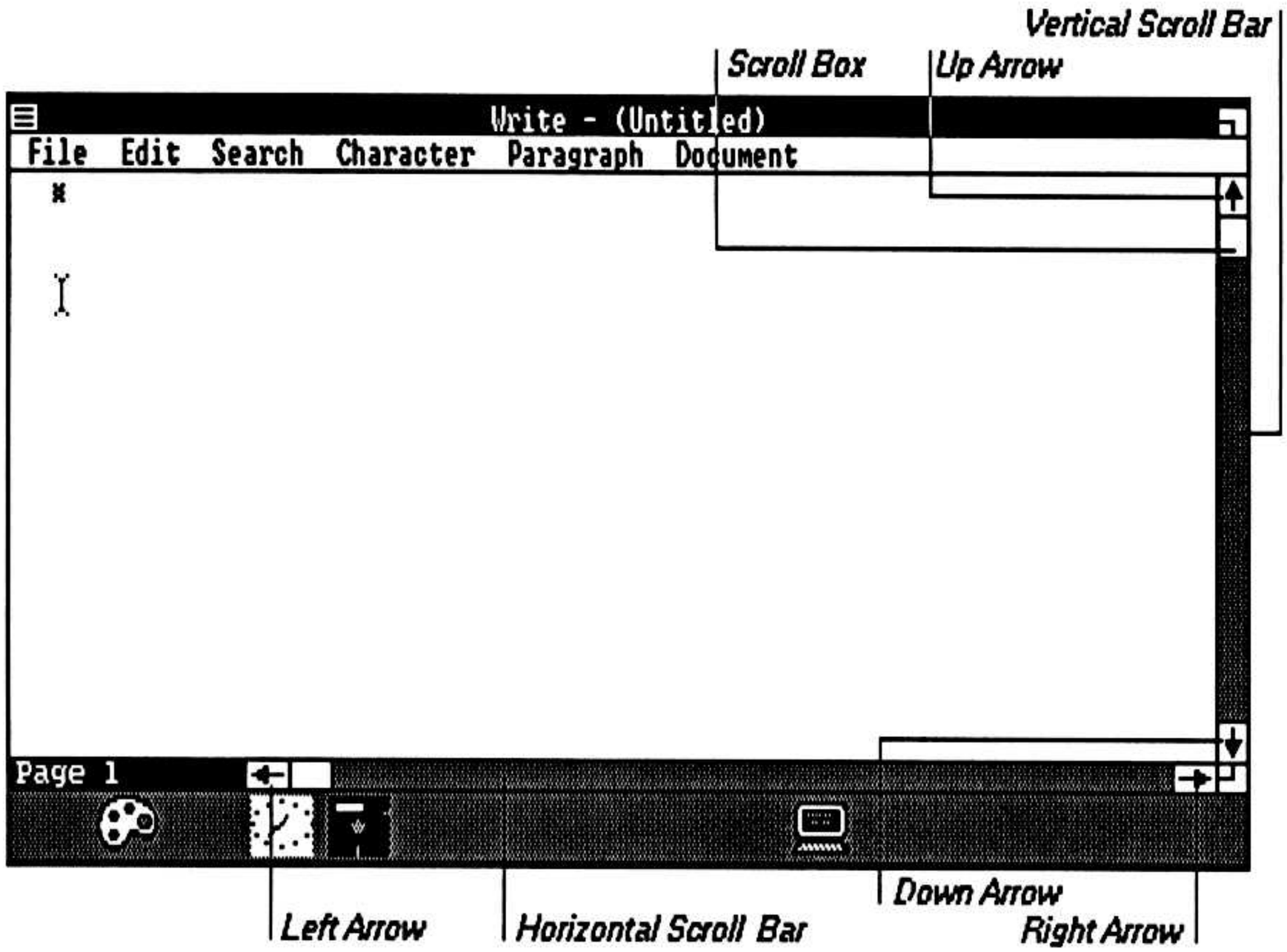


Figure 1.4 Scroll Bars

1.6 Split Windows

Applications can split the client area of a window into separate viewing areas by creating child windows in the client area. A split window permits different types of output within the same main window. An example might be a spread-sheet application, in which the user needs to see different areas of the data simultaneously. To accomplish this, the application can split the screen, creating child windows.

Applications that split windows in this way are responsible for maintaining each individual window. A child window must reside within the client area of its parent window. Child windows do not have icons to represent them.

The following screen shows a split window:

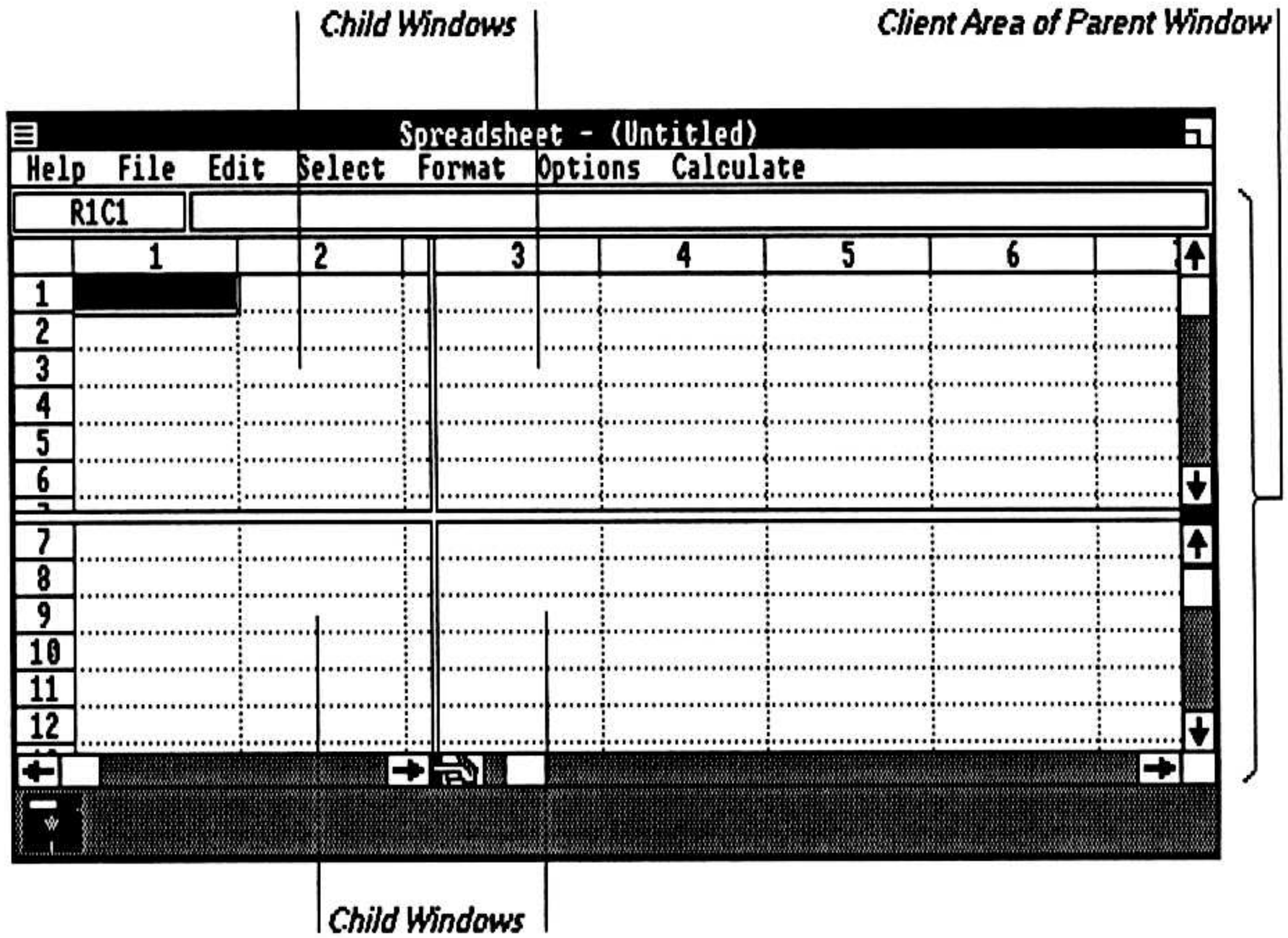


Figure 1.5 Split Window with Four Child Windows

1.7 Popup Windows

An application may also use popup windows, which overlap the existing window. Popup windows may be used to obtain additional information from the user. Examples are the dialog boxes that appear on the screen in response to certain menu selections. Windows provides the functionality for creating dialog boxes for this purpose, but you can create your own popup windows if desired. Whereas a child window must stay within its parent's client area, a popup window may appear anywhere on the screen. Like child windows, popup windows do not have icons.

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Menus

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2.1 Introduction

Applications developed for the Microsoft Windows operating environment that have commands should provide menus to give the user access to the commands. A menu is a list of items, such as commands, from which the user chooses. Each menu is represented by a name in the menu bar of the application's main window. The list of items is displayed when the user selects the corresponding menu name using the keyboard or the mouse.

2.2 Menu Names

Every menu must have a menu name. The menu name represents the complete menu, so it should always state as clearly as possible the purpose of the items in the menu. Compound names may be used (for example, Fontsize).

To give keyboard users easier access to menus, applications should have menu names with unique first letters. With unique letters, a user can access a menu with a single ALT–*letter* key combination. Menu names should not be padded with spaces.

The following screen shows a list of menu names in the menu bar:

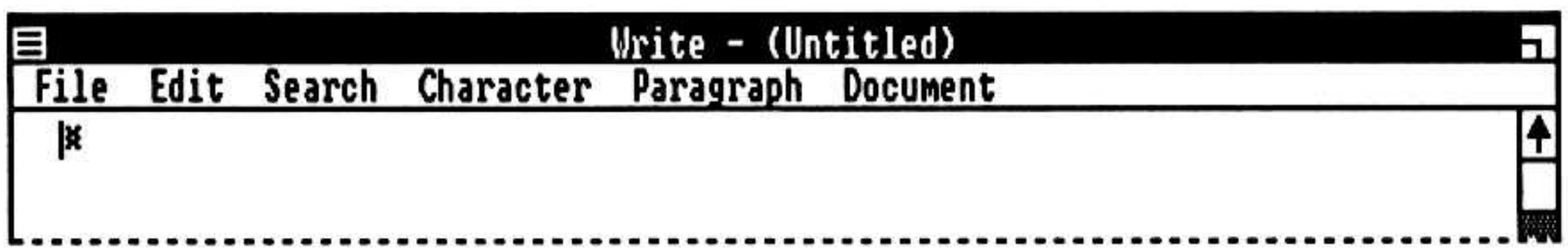


Figure 2.1 Menu Names

2.3 Menu-Item Names

Menu-item names should specify the action the user will initiate. Item names should be unique within a menu, but applications may use the same item name in different menus to represent different actions. Compound names may be used.

In general, items that have similar functions should be placed together. Items that are clearly different should be separated from other items with a line separator. Menu-item names should not be placed in the menu bar. Only menu names should be included in the menu bar. The following figure shows a sample listing of menu items from the Microsoft Write Character menu:

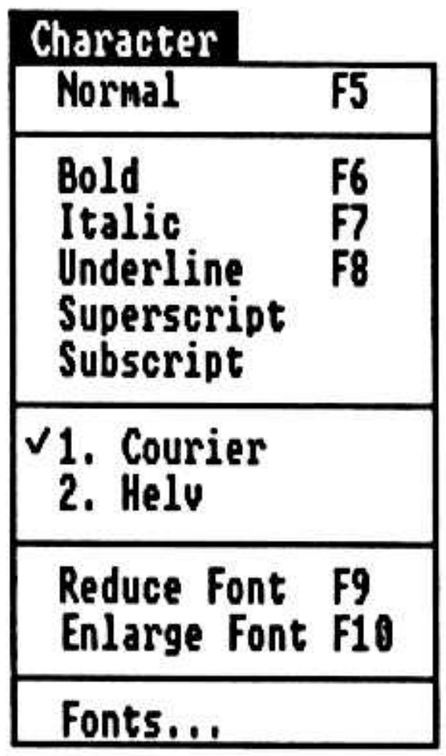


Figure 2.2 Items in a Menu

2.4 Menu-Item Types

Menu items are of several types. They may cause a specific command to be carried out, or they may allow the user to choose from among several features. In other cases, a menu item gives the user the option to turn a certain feature on or off (toggle commands). Some menu items request additional information from the user by generating a dialog box.

Menu items include the following:

Item	Description
Commands	Items which carry out actions, such as opening or deleting a file.
State setters	Items which set some system state. For example, these may be used to turn a feature on or off (toggle commands) or to choose from several features.
Dialog requests	Items which cause dialog boxes to appear. The user then is required to supply additional information, make additional choices, or cancel the request.

The following figure shows several types of menu items:

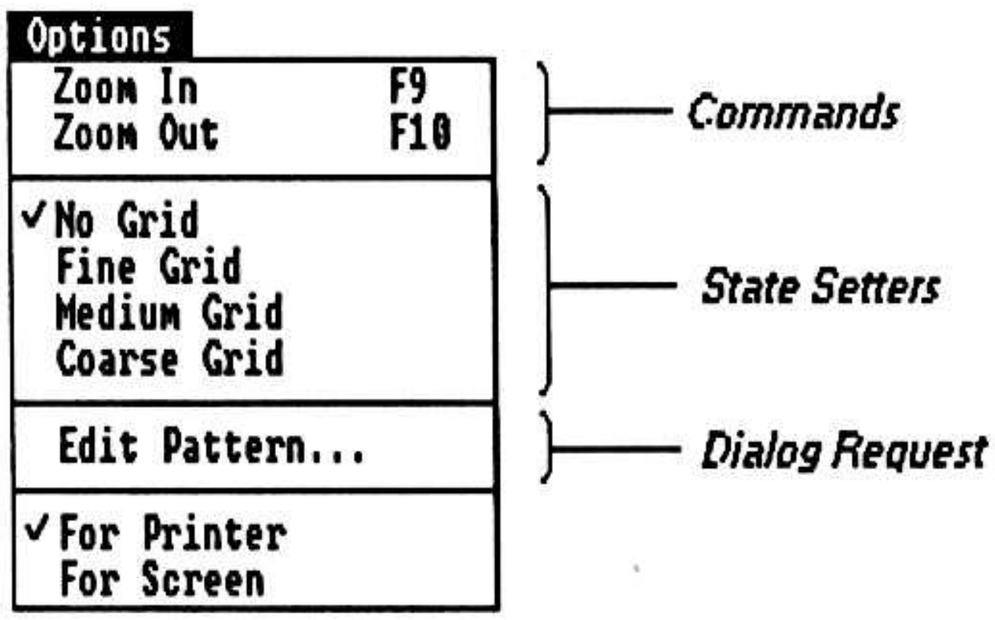


Figure 2.3 Types of Menu Items

2.4.1 Toggle Commands

Applications should always show whether the system state implied by a toggle command is active or not. If the command is on, the application should place a checkmark next to the command name. Otherwise, no checkmark should appear.

Menus may contain any number of toggle commands. Toggle commands that affect the same command should be placed together and should be separated from other commands with a line separator. If many toggle commands are in one group, at least one command should turn

off all commands in that group. If two or more toggle commands are mutually exclusive, the application must make sure that the commands cannot be checkmarked at the same time.

The following figure shows a toggle command with a checkmark to show that the command is active:

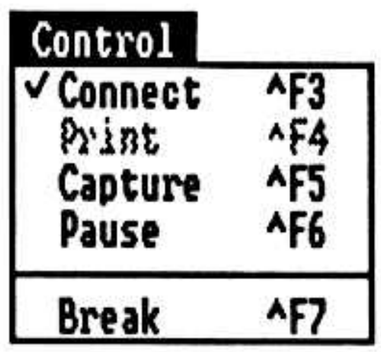


Figure 2.4 Toggle Command in a Menu

2.4.2 Items Using Dialog Boxes

Menu items that need additional information before they can carry out their actions should prompt the user for the information by displaying a dialog box. The dialog box requires action by the user before the application can continue. The menu-item name should end with an ellipsis (...) and the item, when selected, should display a dialog box.

For more information on dialog boxes, see Chapter 8, "Dialog Boxes."

2.5 Disabled Menu Items

The application must determine when the use of a menu item is appropriate. When the use of an item is not appropriate, the application must disable and gray the item. A grayed item lets the user know the item is disabled. If the user attempts to choose a disabled item, no action by the application is necessary. (Due to programming requirements, it is possible for the user, when using the keyboard interface, to *choose* a grayed item. However,the named action will not be carried out.)

The following figure shows a grayed menu item:

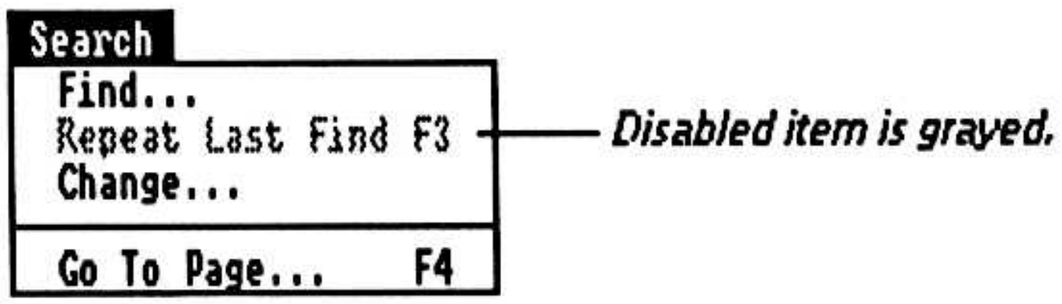


Figure 2.5 Disabled Menu Item

2.6 The System Menu

The System menu is predefined by the Microsoft Windows operating environment. Thus, the list of menu items, the order in which they appear, and their functions are all preset and cannot be changed.

2.6.1 Disabled System Menu Items

When an application is not using items in the System menu, the items should be grayed. For example, an application may not allow its window to change sizes. When this application is running, the Size command listed in the System menu would appear in gray.

2.6.2 Close Command

If the user selects the Close command from the System menu after making changes without saving them, the application should display a warning in a dialog box. (For more information on dialog boxes and warnings, see Chapter 8, "Dialog Boxes.")

2.6.3 About... Command

The only command that may be added to the System menu is the About... command. All applications that wish to provide extra information or help should place the About... command at the end of the System menu of the application's main window and provide help through this command rather than through commands in its own menu bar. (The F1 function key is often used as the Help key.)

When the About... command is appended to the System menu, the command should be separated from the other commands in the menu by a horizontal line.

When the user chooses the About... command, the application should display a dialog box that contains the application name, the date the application was created, copyright information, and any additional information about the application, such as available workspace or the meaning of specific keys.

Every About... dialog box needs an Ok button, shown with a bold border, to allow the user to return to the application. Applications that provide additional help may have other buttons, too.

For more information on dialog boxes, see Chapter 8, "Dialog Boxes."

2.7 The File Menu

Every application that works with data files should have a File menu. The File menu should provide all commands needed to open and create files and to save data in files. The File menu should be the first listed in the menu bar.

It is recommended that the File menu contain the following items and that they have the following actions:

Item	Action
New	Deletes all existing data, removes the current filename, and starts the application from the beginning. If this command could destroy existing data, it should generate a dialog box asking the user to verify the action before it is carried out.
Open...	Prompts the user for the name of the file to be opened, then opens it and uses the file as the current file.
Save	Saves the existing data in the current file. If the file is untitled, the application should generate a Save As... dialog box.

Save As...	Prompts the user for the name of a file to save the current data in, then opens the file and copies data to it.

The following figure shows the recommended format for the File menu:



Figure 2.6 File Menu

Menu items that prompt for user input should use application-modal dialog boxes. (For more information, see Chapter 8, "Dialog Boxes.")

Note

In all cases, applications should take the safe approach when carrying out default actions with files; that is, they should always attempt to prevent the loss of data from existing files.

2.7.1 New and Open... Commands

If the user selects the New or Open... command (or the Close command in the System menu) after making changes without saving them in a file, the application should display a dialog box with the following format:

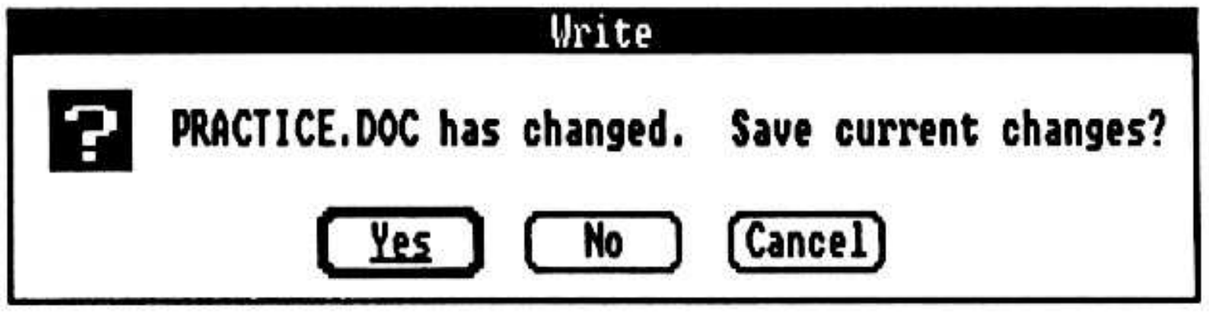


Figure 2.7 Dialog Box with Warning

The application should use the appropriate description of the data file in the first line of the message. The Yes button is the default button.

2.7.2 Save Command

The Save command should overwrite the existing file without asking permission, but if there is no current filename, the application should carry out a Save As... command instead.

2.7.3 Save As... Command

The Save As... command should display a dialog box with the following format:

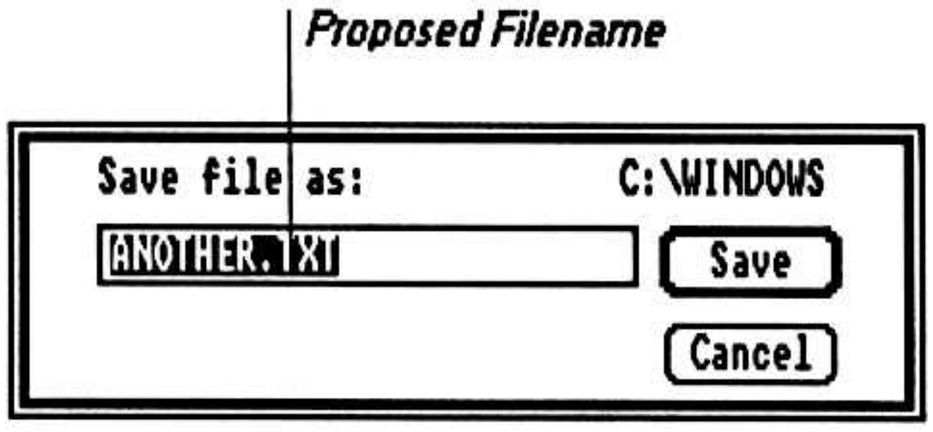


Figure 2.8 Save As... Dialog Box

The Save As... dialog box should propose the existing filename as the default. If there is no existing filename, it should propose a reasonable default. The Save button is the default button.

If the command would overwrite an existing file, the Save As... command should always display a dialog box to ask permission to overwrite, as shown in the following example:

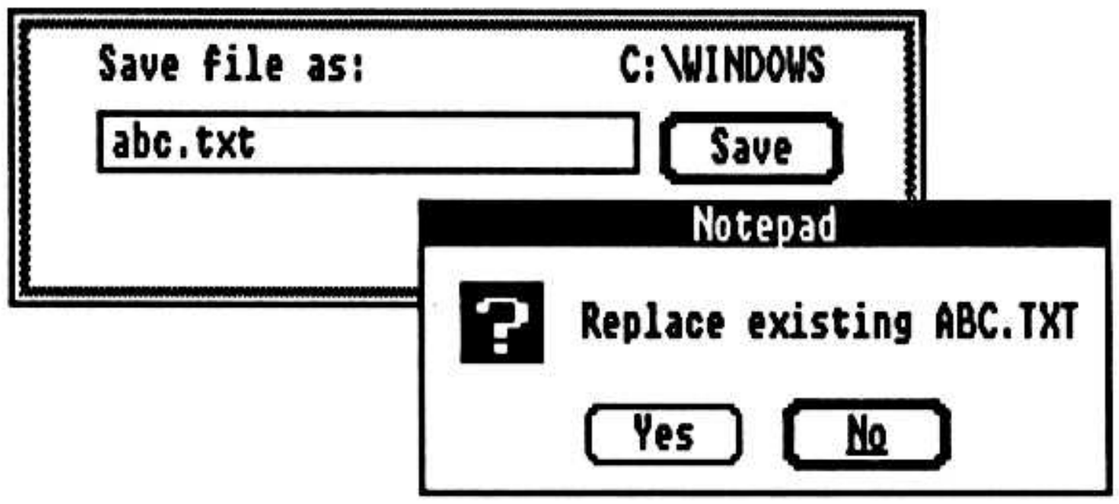


Figure 2.9 Save As... Dialog Box with Warning

If the user selects the No button, the application should return to the Save As... dialog box. Otherwise, the application should attempt to save the file. The No button is the default button.

2.8 The Edit Menu

Every application that lets the user copy text or graphics to the Clipboard should have an Edit menu. The Edit menu should provide all the commands needed to carry out Clipboard operations.

The Edit menu should contain at least the following commands:

Command	Action
Undo	At a minimum, reverses the action of the last Clipboard command. You may wish to add more information and functionality to this command.
Cut	Copies the current selection to the Clipboard, then deletes the selection.
Copy	Copies the current selection to the Clipboard without deleting.
Paste	Copies the contents of the Clipboard into the application's data file.
Clear	Deletes the current selection but leaves the Clipboard unchanged.

Commands that require a selection should create a selection using the techniques described in Chapter 4, "Selection."

If no current selection exists, the Cut, Copy, and Clear commands should be disabled. If the Clipboard is empty, the Paste command should be disabled. If there is no previous Clipboard command, the Undo command should be disabled. The following figure shows disabled Edit-menu commands:

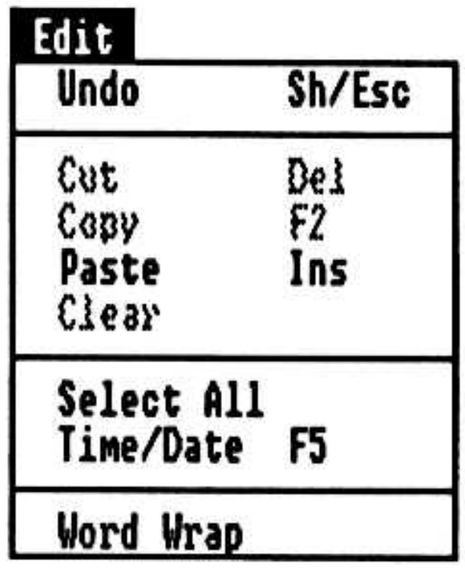


Figure 2.10 Edit Menu with Disabled Commands

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Accelerators

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3.1 Introduction

Accelerators (shortcut keys) are keystrokes that let the user carry out the action of a menu item more easily than by the normal method of selecting the item. An accelerator is typically a CONTROL–*key* combination or a function key that the application has explicitly defined to represent a menu item. Pressing the key carries out the item just as if the user had displayed the menu and selected the item.

3.2 Accelerator Rules

Applications developed for the Microsoft Windows operating environment can provide any number of accelerators but should restrict their use to the menu items used most often. An item should never have more than one accelerator.

3.3 Accelerator Names in Menus

Applications that use accelerators should place the accelerator's keyname beside the corresponding menu-item name in the menu. The keyname should appear to the right of the item name.

If an accelerator is a CONTROL–*key* sequence, it should be shown as ^*key*, where *key* is the keyname. There should be no space between the caret and the keyname. If an accelerator is a function key, it should be shown as F*n* where F is uppercase and *n* is the digit. There should be no space between the letter and digit.

The following figure shows accelerator keynames listed to the right of the menu items:

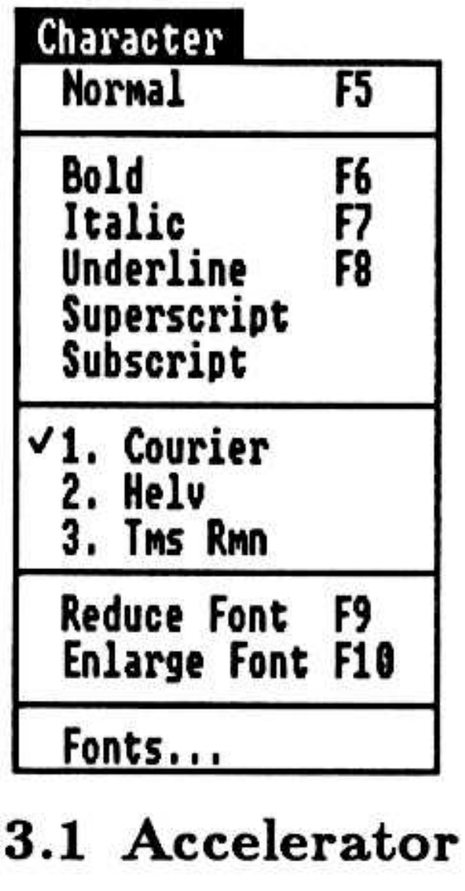


Figure 3.1 Accelerator Names

3.4 Standard Edit-Menu Accelerators

Applications that use accelerators with Edit-menu commands should use the following keys:

Key	Action
SHIFT-ESCAPE	Invokes the Undo command.
DELETE	Invokes the Cut command.
F2	Invokes the Copy command.
INSERT	Invokes the Paste command.
SHIFT-DELETE	Invokes the Clear command.

Applications should ignore the SHIFT-DELETE, DELETE, and F2 Edit-menu accelerators if there is no current selection. They should ignore INSERT if the Clipboard is empty and ignore SHIFT-ESCAPE if there is no previous Edit-menu command.

In addition to these standard Edit-menu accelerators, F1 is often reserved as the Help key. The user presses F1 to obtain more information about a subject.

Note

If an application provides an Edit menu, it must not use the standard Edit-menu accelerators for any other purpose.

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4.1 Introduction

Applications developed for the Microsoft Windows operating environment that let the user copy information to the Clipboard or that have commands that act on the contents of the screen should provide a way to let the user select the information to be copied or acted on. The selection is one or more items that the user has chosen. These items will be used as input to a subsequent command.

There are two kinds of selections: text and non-text items. Text selections are such things as a word or phrase in a manuscript the user is writing. Non-text selections can be either an item, such as a cell on a spread sheet, or a region on the screen, such as part of a drawing the user is making.

Applications that provide selection should let the user select with either the keyboard or the mouse. Applications should use the keys and mouse buttons recommended in Chapter 5, "Keyboard Interface," and Chapter 6, "Mouse Interface."

4.2 Selection Highlight

Applications should mark the current selection by highlighting it. The highlight can be acheived by displaying the selection in reverse video (used for text and item selections) or by enclosing it in a rectangle (used for some graphic selections).

When the window containing the selection is no longer active, the application should remove the highlight. When the window becomes active again, the application should restore the highlight.

If the window containing the selection is no longer active but the application is still using the selection in some way, the application should keep the selection highlighted. For example, in Microsoft Windows Write, if the user selects a section of text, then chooses Fonts from the Character menu, a dialog box appears. The dialog box becomes the active window, but the text remains selected, as shown in the following screen:

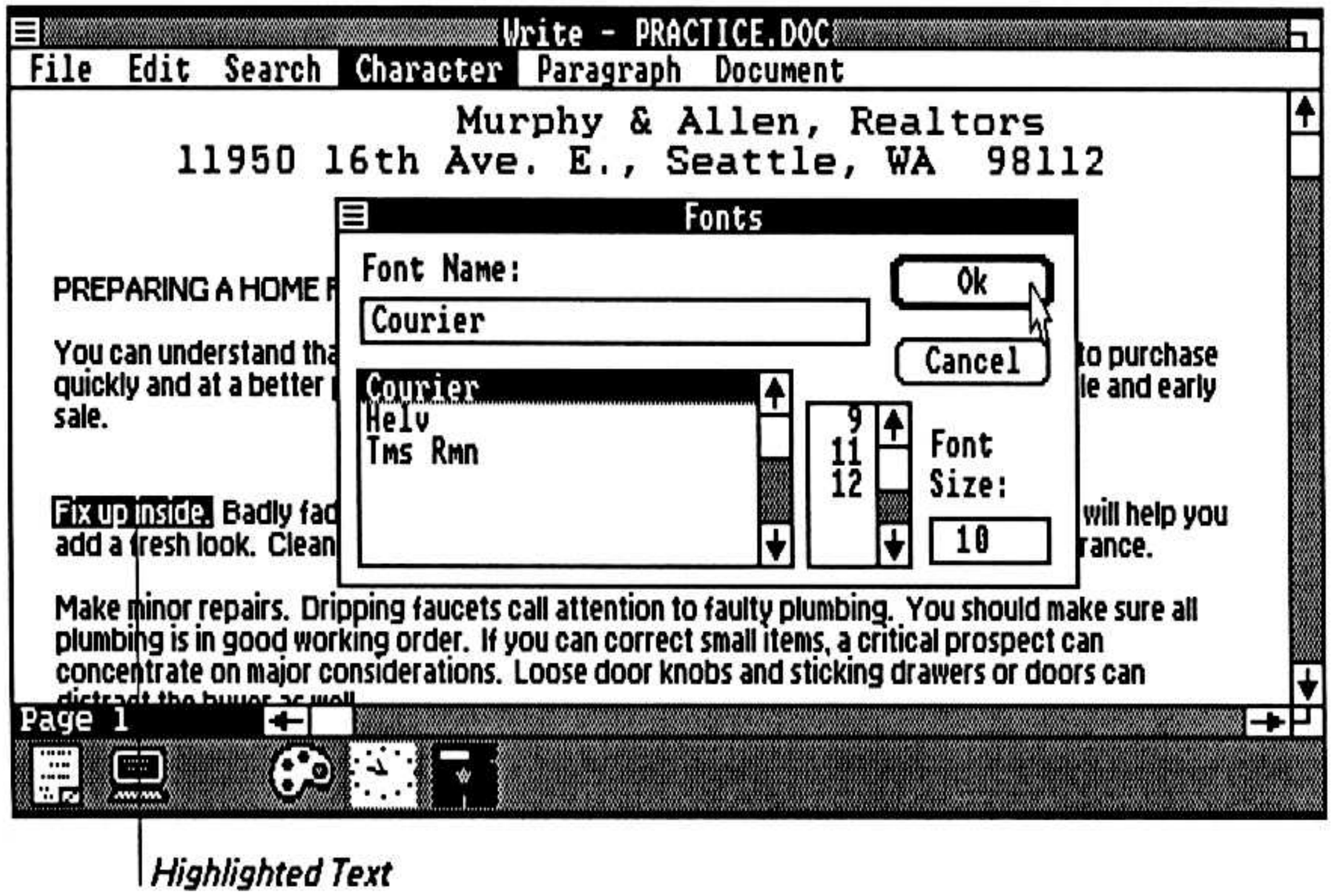


Figure 4.1 Selected Text

4.3 Text Selection

When an application permits the user to select text, the following types of selections are recommended:

Selection	Result
Insertion point	User should be able to select a point between characters to insert new characters.
Character	User should be able to select individual characters.
Word	User should be able to select individual words. A word is any contiguous string of characters not containing a word-break character. Word-break characters are such things as new-line characters and tabs. Punctuation can be selected the same as a word.

Line	User should be able to select whole lines. A line is any contiguous string of characters not containing a new-line character.
Extended	User should be able to extend the existing selection to include additional characters. For example, the user can extend a character or word selection to several more characters.

All text selections, except the insertion-point selection, should be highlighted. The insertion-point selection should be marked with a flashing caret (shown as a flashing bar or bitmap).

An extended selection should always begin at the original selection and continue either to the left or to the right (or up or down). An application should not allow a selection to be extended in both directions at the same time. Instead, there should be one active edge of selection. If, when using the keyboard, the user extends the selection to the beginning or end of a line, the extension should move to the opposite end of the next line up (if moving left) or down (if moving right). Extending the selection up is the same as extending it to the left quickly; extending it down is the same as extending it to the right quickly. If, while extending the selection to the left (or right), the user moves the selection in the opposite direction, the application should remove the extended selection in the reverse order from which it was made.

The application should let the user remove a text selection by making some other selection.

4.4 Editing Text Selections

Applications that permit editing of the currently selected text should follow these rules:

- When text is inserted at an insertion point, the insertion point moves to the right and new characters are placed to the left of the point.
- When a character, word, or line is selected, inserting text removes the selected text and replaces the selection with an insertion point. The insertion continues as described above.

- When a character, word, or line is selected, deleting text removes the selected text.
- Using the BACKSPACE key at an insertion point deletes the character to the left and moves the insertion point to the left.
- Using the SHIFT-BACKSPACE key sequence at an insertion point deletes the character to the right of the insertion point.

4.5 Non-text Selection

Applications that permit the user to select items other than text should provide the following type of selections:

Selection	Result
Item	User should be able to select individual items on the screen, such as filenames or icons.
Region	User should be able to select the contents of a rectangular region on the screen.

Applications should let the user make any number of item selections. The selections should be highlighted by displaying them in reverse video or bordered by handles. Extended selection is not permitted; instead, the user can expand the selection by selecting additional items, as shown in the following figure:

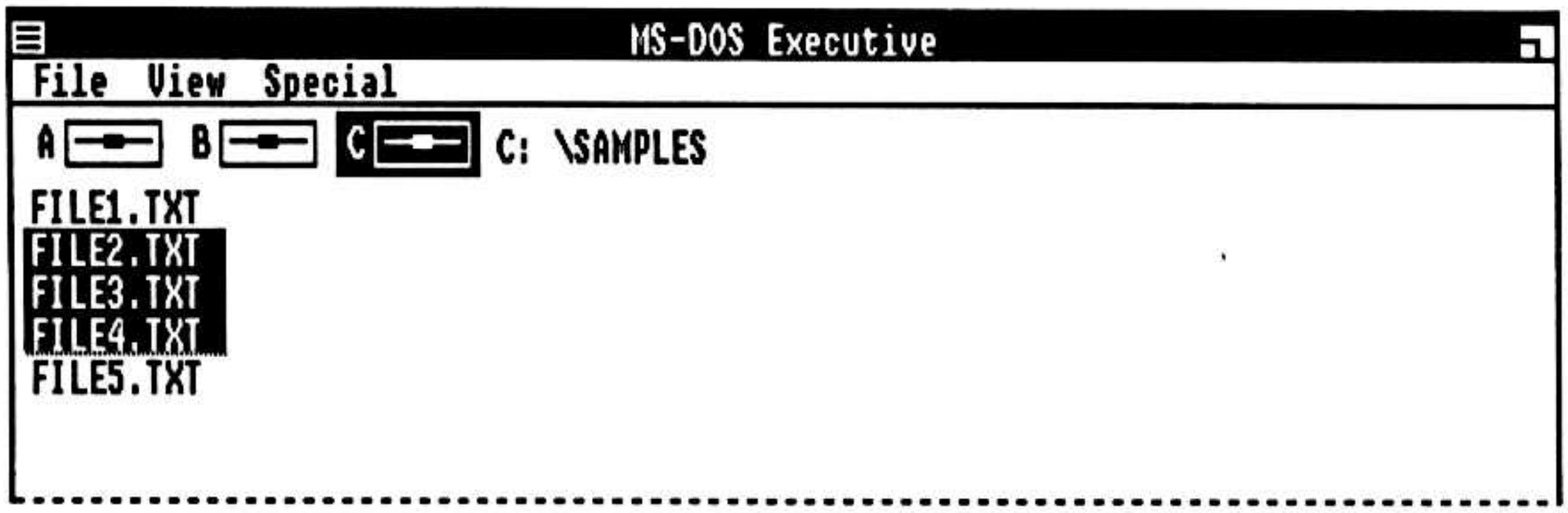


Figure 4.2 Selected Items in Reverse Video

Applications should let the user select any rectangular region on the screen. The user begins the selection at any point and moves it any direction to an end point. The start and end points mark the diagonally opposite corners of the rectangular region. While the user selects the region, the application should highlight the region to mark the user's progress. The highlight can be a reverse-video display or an enclosing rectangle. A selected region is shown highlighted with an enclosing rectangle in the following screen:

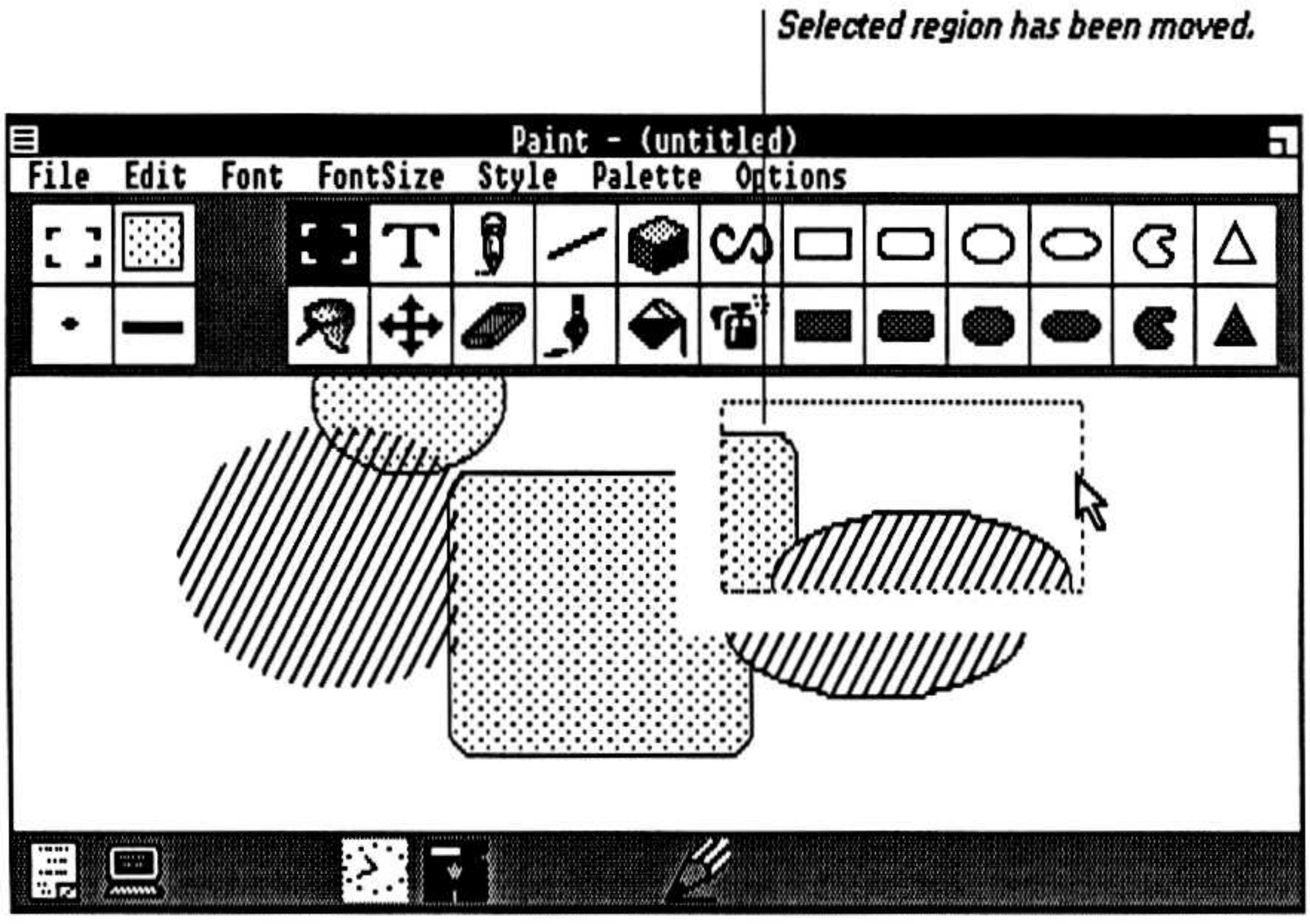


Figure 4.3 Region Selected with Rectangle

4.6 Discontinuous Selection

A discontinuous selection is any non-text selection in which two or more non-adjacent items on the screen are highlighted. An application can permit discontinuous selections whenever the user needs to select multiple items from a list of items, such as two or more filenames from a directory listing.

Applications should not permit discontinuous text selections. Instead, each new selection should clear the previous selection.

Chapter 5
Keyboard Interface

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5.1 Introduction

Since the Microsoft Windows operating environment does not require a mouse or similar pointing device, applications should always ensure that every menu item can be selected using the keyboard. (To test this, it is recommended that you unplug the mouse card and try every menu item with the keyboard.)

To assist application design, Windows provides a built-in keyboard interface for accessing an application's menus; for selecting the active application, window, or control; and for moving, sizing, zooming, and closing a window. For other actions, such as scrolling a window and selecting objects in a window, the use of specific keystrokes is recommended, but the application should carry out the actual operations.

5.2 Reserved System Keys

The reserved system keys are the built-in system keys used by Windows to access an application's menus and dialog boxes. Applications can use some of these keys for normal input and output when the application has the input focus. *Applications must never attempt to trap these keys.*

5.2.1 Do Not Redefine

The following reserved keys are active at all times and must not be redefined by any application:

Key	Action
ALT-TAB	Selects active window, moving from top to bottom and left to right.
ALT-SHIFT-TAB	Selects active window, moving from bottom to top and right to left.
ALT-SPACEBAR	Selects the System menu of the active window.
ALT-ENTER	Zooms or dezooms the current window.
ALT–*letter*	Selects a menu from the active window's menu bar. The *letter* is the first letter of the menu name.

5.2.2 Cannot be Redefined While Menu in Use

The following reserved keys are active when a menu has been selected. At other times, they may be redefined by an application.

Key	Action
LEFT-DIRECTION	Selects menu to the left of the current menu.
RIGHT-DIRECTION	Selects menu to the right of the current menu.
UP-DIRECTION	Selects command above the currently selected command in the selected menu.
DOWN-DIRECTION	Selects command below the currently selected command in the selected menu.
ENTER	Carries out selected command.
ESCAPE	Cancels menu.

5.2.3 Cannot be Redefined While Dialog Box in Use

The following reserved keys are active when a dialog box is displayed. At other times, they may be redefined by an application:

Key	Action
TAB	Selects the active control, moving from left to right and top to bottom.
SHIFT-TAB	Selects the active control, moving from right to left and bottom to top.
SPACEBAR	Activates the selected button.
ENTER	Carries out the default action as defined by the default push button.
ESCAPE	Cancels the dialog box.

5.3 Recommended Keyboard Use

Although applications can use all keys on the keyboard for their normal input operations, for the sake of a consistent keyboard interface, it is recommended that applications use the following keys for the actions listed.

5.3.1 Keys Used for Selecting Items

Key	Action
TAB	Moves to next item from left to right and top to bottom.
SHIFT-TAB	Moves to previous item from right to left and bottom to top.
ENTER	Confirms or executes the default item.
LEFT-DIRECTION	Moves left one item.
RIGHT-DIRECTION	Moves right one item.
UP-DIRECTION	Moves up one item.
DOWN-DIRECTION	Moves down one item.
PAGE UP	Moves up one screen.
PAGE DOWN	Moves down one screen.
INSERT	Inserts contents of Clipboard at cursor.
DELETE	Moves selected item to Clipboard.
SHIFT-UP-DIRECTION	Extends selection up from current item.
SHIFT-DOWN-DIRECTION	Extends selection down from current item.
SHIFT-LEFT-DIRECTION	Extends selection left from current item.
SHIFT-RIGHT-DIRECTION	Extends selection right from current item.
SHIFT-DELETE	Deletes selected item.
CONTROL-*direction*	Leaves the selection intact; moves the underscore in the indicated direction.

CONTROL-SHIFT-*direction*	Adds to a discontinuous selection, moving in the indicated direction.

5.3.2 Keys Used for Selecting Text/Graphics

Key	Action
TAB	Moves text one tab stop to the right.
ENTER	Creates an empty line.
LEFT-DIRECTION	Moves left one character.
RIGHT-DIRECTION	Moves right one character.
UP-DIRECTION	Moves up one line.
DOWN-DIRECTION	Moves down one line.
PAGE UP	Moves up one screen.
PAGE DOWN	Moves down one screen.
SHIFT-UP-DIRECTION	Extends selection up from current selection.
SHIFT-DOWN-DIRECTION	Extends selection down from current selection.
SHIFT-LEFT-DIRECTION	Extends selection left from current selection.
SHIFT-RIGHT-DIRECTION	Extends selection right from current selection.
SHIFT-BACKSPACE	Deletes character or space to right of insertion point.
BACKSPACE	Deletes character or space to left of insertion point.
SHIFT-DELETE	Deletes selected item.
HOME	Moves to top of current screen.
END	Moves to end of current screen.

5.4 Input Focus

An application should acquire the input focus (current ownership of the keyboard) whenever it is the active window. Since only one window can have the input focus at a time, having the focus guarantees that the application receives all input intended for it.

An application should relinquish the input focus whenever the user moves to another window. This permits the new window to acquire the focus if necessary. The application should reacquire the focus when the user returns to the application's window.

Although Windows provides a mechanism to trap keyboard input (and circumvent the input focus), applications should use this mechanism only when absolutely necessary.

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Mouse Interface

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6.4 Mouse Capture 51

6.1 Introduction

Although a mouse (or similar pointing device) is optional with the Microsoft Windows operating environment, applications should use it liberally as an alternate method of input. In particular, applications should always use the mouse wherever it clearly improves the application's performance.

Note

Since the mouse is the most common pointing device, for convenience, the word "mouse" is used in all discussions of pointing devices. However, information about using a mouse pertains to other pointing devices as well.

6.2 Recommended Mouse Use

As with the keyboard, Windows provides a built-in mouse interface for accessing an application's menus; for selecting the active application, window, or control; and for moving, sizing, zooming, and closing a window. Other specific actions should correspond to the mouse buttons and movements listed below. The application is responsible for carrying out the actual operations.

The left mouse button should be used for all of the application's actions. Other buttons may be used for shortcuts, but should not be required by the application. Applications should use the mouse as shown in the following lists.

6.2.1 Text Selection with a Mouse

Mouse Movement	Action
Click	Moves insertion point to location being pointed to.
Double-click	Selects a word.

Drag	Extends selection from the location being pointed to through to the release point.
SHIFT and Click	Extends selection from the last selected character to the pointer. For a left-to-right selection, removes the selection that follows the location being pointed to. For a right-to-left selection, removes the selection that precedes the location being pointed to.
SHIFT and Drag	Extends selection from the end of the current selection through to the release point. Removes selection from the location being pointed to through to the release point.

6.2.2 Item Selection with a Mouse

Mouse Movement	Action
Click	Selects item being pointed to.
Double-click	Confirms or executes item being pointed to.
Drag	Extends selection from the location being pointed to through to the release point.
SHIFT and Click	Extends selection from previous end point to the location being pointed to.
CONTROL-Drag	Allows a discontinuous selection: keeps current selection and creates an additional selection from the location being pointed to through to the release point.
CONTROL-Click	Serves as a toggle command: by using CONTROL-Click, the selection is deleted or restored.

6.3 Mouse Cursor

Applications should use mouse cursors that are appropriate for the current action. When the user needs to select items by pointing with the mouse, the application should provide a pointer. A variety of pointers may be needed for the same action when the cursor is in different parts of the window.

Three mouse pointers are shown as follows:

Pointer	Description
	Arrow pointer. Used to select menus, items, commands, etc.
	I-beam pointer. Used to indicate an insertion point.
	Hourglass pointer. Indicates that an operation is taking place although no output is being sent to the screen.

6.4 Mouse Capture

Under normal operating conditions, an application receives mouse input only when the mouse cursor is in the application's window. Applications that wish to receive mouse input from points outside the window can capture the mouse input, but should do so sparingly. When mouse input is captured, the user cannot select menus nor change to another application.

Chapter 7
Filenames

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7.1 Introduction

Applications developed for the Microsoft Windows operating environment that use filenames to create and open files should use a consistent approach when forming filenames from user input.

7.2 Default Filename Extension

Applications should use the same extension for all data files whenever possible. In particular, an application should have a default filename extension that it appends to a data filename whenever the user does not supply one. The default extension should be unique so that the user can distinguish one application's data files from another or use the file to invoke the application implicitly. The extension can be one, two, or three characters long.

7.3 Filename Conventions

Applications should apply the following rules when forming complete filenames from user input:

- If the user supplies a filename without a pathname, by default, the application should search the current directory. If the file is not found, the PATH should be searched.

  If the user is saving a file and the file is not found in the search described above, it should be saved in the current directory. In some cases, a Save As... dialog box will be generated. If the file is in the current directory, the application should show only the filename. If the file is in another directory, the complete pathname should be shown.

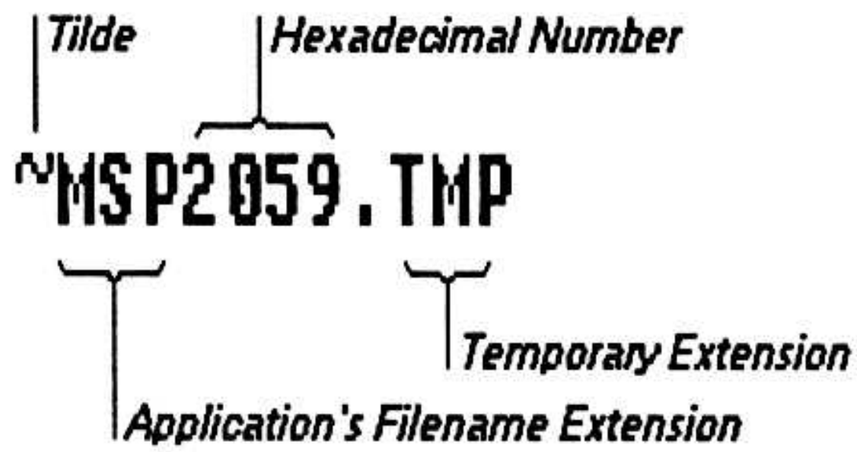
- If the user supplies a pathname, the application should decide if it is appropriate. If it is not, the application should display an error message and prompt for a new name.

- If the user supplies a name without an extension, the application should append the application's default extension.

- If the user supplies a null extension (for example, *foo.*), the application must not append the default extension.
- If the filename is too long, the application should generate an appropriate error message.

7.4 Temporary Filenames

Applications that create temporary files (files to be used during execution only), should use names as shown in the following example:



The ~ is a required tilde.

The *filename extension* is the 1-, 2-, or 3-character default extension used by the application.

The *hexadecimal number* is unique to each new temporary file. This number is created by the **GetTempFileName** function.

The temporary extension *.tmp* is required.

Since it is possible to run multiple copies of an application simultaneously, each temporary file must have a unique name to prevent one file from overwriting another. The Windows **GetTempFileName** function can be used to assign a unique name to each file.

Temporary files should always be placed in the directory specified by the user's DOS TEMP environment variable, if one is specified. If no directory is specified, the files should be placed in the *root* directory.

7.5 Prompting for Directories and Filenames

Commands that require a filename or directory name to carry out their actions should prompt for the names using a dialog box with the following format:

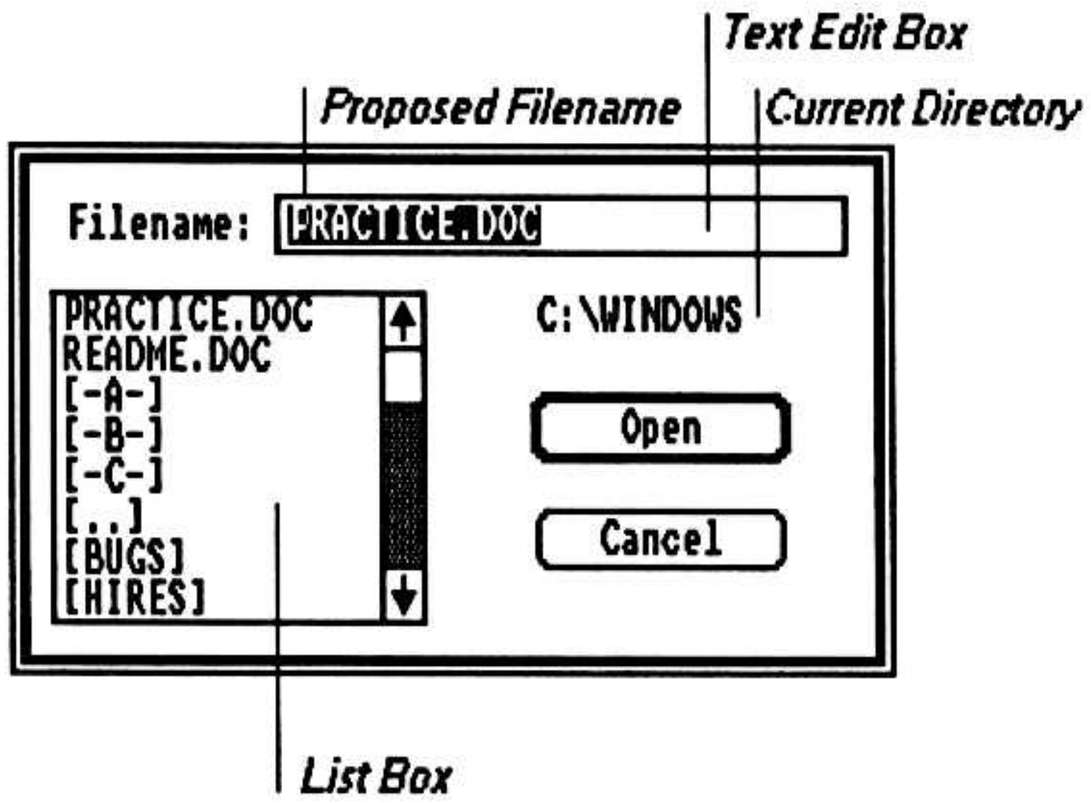


Figure 7.1 Dialog Box Requesting a Filename

The text-edit box is for entering the desired file or directory name. The list box is for listing the contents of a directory.

The application should place the current filename in the text-edit box as the proposed filename. If there is no current file, the application should propose an appropriate default name. The application should also select the proposed name for editing. This lets the user delete the name with a single keystroke if another name is desired.

The application should list the partial contents of the current directory in the list box. In particular, it should list all directory names and all filenames that end with the application's extension. Whenever the user selects a filename from the list box, the application should copy this name to the text-edit box.

For more information on using list boxes, see the *Microsoft Windows Programmer's Reference.*

Chapter 8
Dialog Boxes

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8.1 Introduction

Applications developed for the Microsoft Windows operating environment should use dialog boxes to request input from the user and to display information that is not part of the application's normal output. A special form of dialog box, called the message box, is used to display error messages and warnings. Message boxes are described in Chapter 9, "Message Boxes."

8.2 Dialog-Box Types

Dialog boxes can be modal or modeless. A modal dialog box prevents the user from continuing to work with the application until some action is taken. A modeless dialog box lets the user continue working with the application.

Applications should use modal dialog boxes for requesting command input. In this way, a modal dialog box ensures that the user responds to the input request before continuing to work.

Applications should use modeless dialog boxes for displaying information that does not require immediate attention. Modeless dialog boxes are typically used to display information that the user can use while working with the application. Unlike modal dialog boxes, modeless dialog boxes can be moved out of the way by the user, so the user's work is never completely obscured.

There are three types of dialog boxes:

Dialog Box	Description
Modeless	This dialog box is a separately functioning window. The user can continue to work with the application that called it or with any other application. The user can move the modeless dialog box if it obscures the work area. The user can close it at any time by using the System menu.

Application modal	This dialog box requires attention before the user can continue to use the application that displayed the box. This dialog box does not prevent the user from continuing to work in other applications.
System modal	This dialog box requires attention before the user can continue with *any* application. Use only when absolutely needed, such as when warning about an impending fatal system error or unrecoverable error that can disrupt system operation.

8.3 Dialog-Box Appearance

The appearance of an application's dialog boxes should be defined by the following rules:

- Dialog boxes should not contain menu bars.
- Dialog boxes should disable and gray (or remove) all unused System menu commands.
- A modal dialog box should have a framed border.
- A modeless dialog box should have a title bar and a System menu.
- A modeless dialog box should have a caption in its title bar. The caption should be the name of the command that created the box.
- A modeless dialog box should disable and gray (or remove) all System menu commands except Move and Close.

8.4 Controls

Dialog boxes generally contain controls through which the user can supply information for the corresponding menu item. The controls can be any combination of the following:

Control	Description
Push buttons	Command buttons, labeled to indicate what they do (for example, Ok, Cancel).
Radio buttons	Group of option buttons from which the user may select one option.
Check boxes	Options that are on or off.
List box	Box listing available choices.
Text-edit box	Box used for entering information.

The following dialog box shows several types of controls:

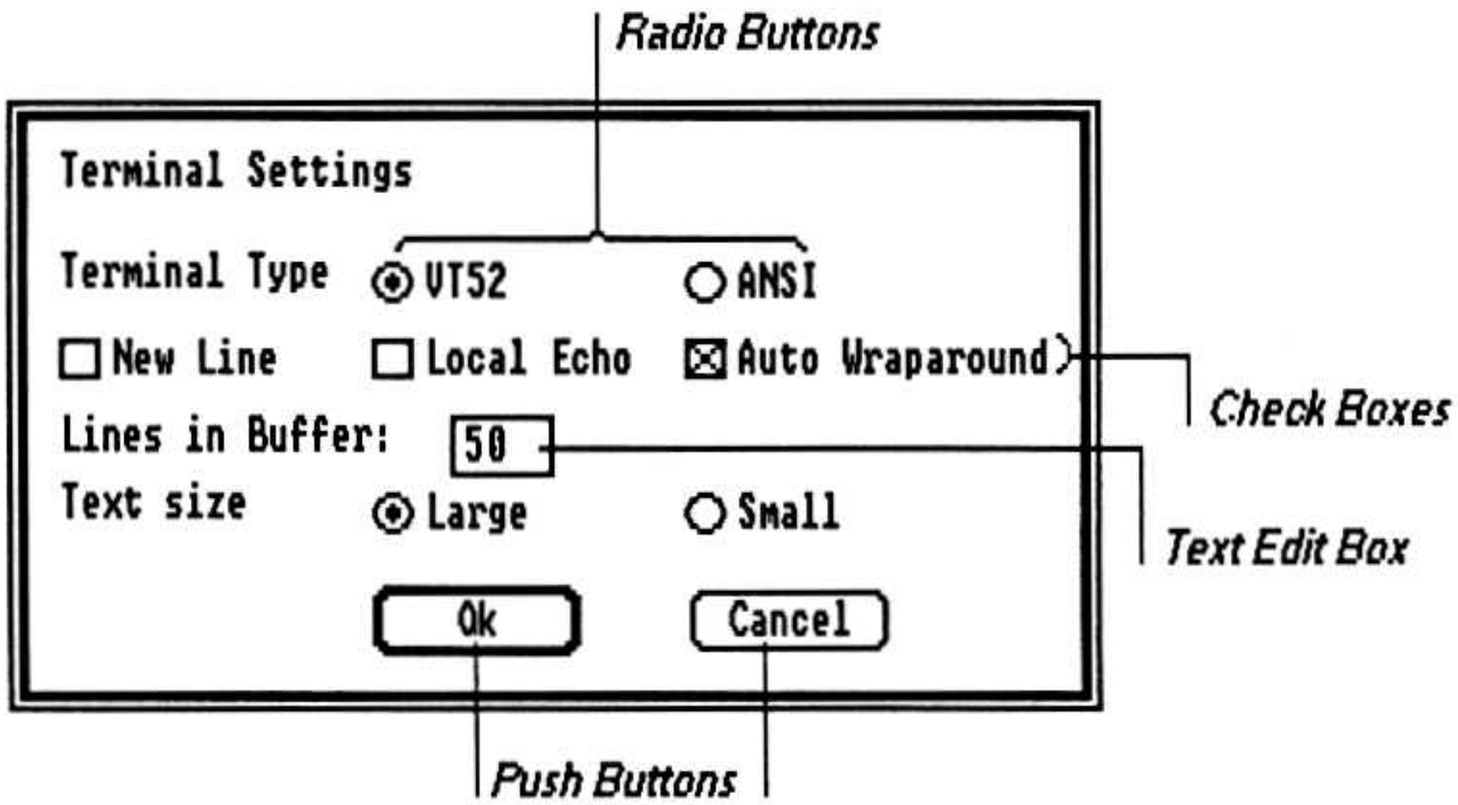


Figure 8.1 Controls in a Dialog Box

The number and type of controls depend on the dialog-box type and the input needs of the corresponding menu item.

The application should always provide proposed values for the information requested in the dialog box and give the user the option of selecting the proposed values or entering new values. Proposed values can be based on previous information or on standard defaults.

Dialog boxes may have a System menu instead of controls, or they may have a System menu in addition to controls.

8.5 Default Buttons

Every modal dialog box should have a default push button with which the user can provide the most reasonable response to the dialog box. The default button has a bold border. For most dialog boxes, the default button carries out the command. In others, the button cancels the command.

In all cases, the ENTER key should activate the default push button, and the SPACEBAR should activate the selected button. The ESCAPE key should cancel the dialog box (if the dialog box has a Cancel button).

8.6 Required Buttons

Dialog boxes that request input should provide at least one push button to carry out the requested action and one to cancel it. In most dialog boxes, the Ok and Cancel buttons are used. The Ok button (the default button) directs the menu item to continue its action with the information supplied. The Cancel button is used to cancel the menu item. In cases where the default or suggested choice is negative, Yes and No buttons are generally used, with the No button highlighted. This avoids any confusion from having the Cancel button be the default in this case. The recommended position for these push buttons is at the upper right edge of the window with Cancel under Ok.

When a button cannot be chosen, the label in the button should be grayed.

8.7 Multiple Buttons

When a push button is activated, its related action is carried out. When check boxes and radio buttons are activated, they change their state (from selected to not selected, or vice versa). The TAB key moves the cursor from one section of the dialog box to another. When a dialog box contains multiple buttons, the SPACEBAR should activate the currently selected button.

8.8 Dialog Boxes without Buttons

It is recommended that all dialog boxes have controls, although some may not. Applications that use dialog boxes without buttons should enable the ENTER and ESCAPE keys to let the user carry out default actions. The application should interpret the ENTER key as confirming, or where appropriate as executing, the actions in the dialog box. Once the action is confirmed, the dialog box should be closed. The application should interpret the ESCAPE key as canceling the dialog box without carrying out an action. In modeless dialog boxes, the ESCAPE key closes the dialog box.

8.9 Sizing Dialog Boxes

Dialog boxes must be sized to fit on all screens the application will run on. Calculate the maximum size of a dialog box as follows:

- Width:

  [(screen width)/(width of system font)]*4=maximum dialog units
- Height:

  [(screen height)/(height of system font)]*8=maximum dialog units

For device-independent applications, the recommended system font is 80×24 characters. The corresponding dialog boxes would be a maximum of 320×96 dialog units. It is recommended that a dialog box take up no more than 75% of the screen.

8.10 Moving between Controls

The normal order of movement within a dialog box should be from top to bottom and from left to right. The following table describes the ways to move within a dialog box:

To	Press
Move within the dialog box	TAB or SHIFT-TAB
Move the insertion point within a text-edit box	DIRECTION keys
Move within a group of check boxes	DIRECTION keys, TAB, or SHIFT-TAB
Move and select within a group of option buttons	DIRECTION keys
Move and select within a list box	UP- or DOWN-DIRECTION keys

8.11 Style Suggestions

This section contains suggestions for design and text format that you may want to use in creating dialog boxes for your applications. Following these guidelines will help ensure a consistent user interface.

8.11.1 Push-Button Design

Push-button height should be 1-1/2 times as high as the tallest character in the button name and also should include space for the blinking underscore.

The first letter in the push-button name should be a capital. All other characters in the name should be lowercase.

8.11.2 Control Appearance

Controls in a dialog box should be placed as follows:

- Place horizontal scroll bars below the window to be scrolled.
- Place vertical scroll bars to the right of the window to be scrolled.
- Label edit controls with static text.
- Related radio buttons should be grouped in one of two ways. Use WS_GROUP to set the style for the first and last buttons in a group. This allows the user to move within the group using the DIRECTION keys. The second method is to enclose the related radio buttons in a group box. A group box is a rectangle that groups controls together.

Examples of both methods are shown in the following figure:

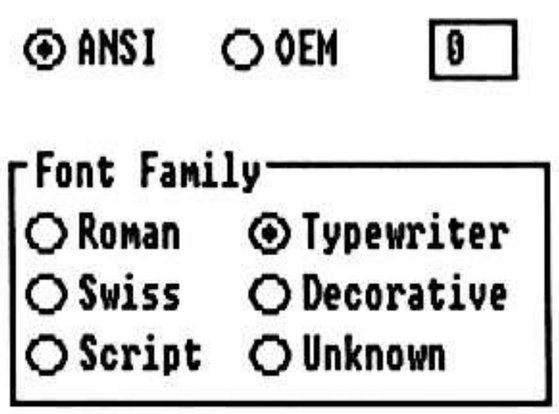


Figure 8.2 Related Radio Buttons

8.11.3 Labels on Controls

Control labels should begin with a capital letter. All text-edit controls should have a static text label that ends with a colon.

Labels on radio button groups should not contain a colon.

The following dialog box shows the recommended format for labels on controls:

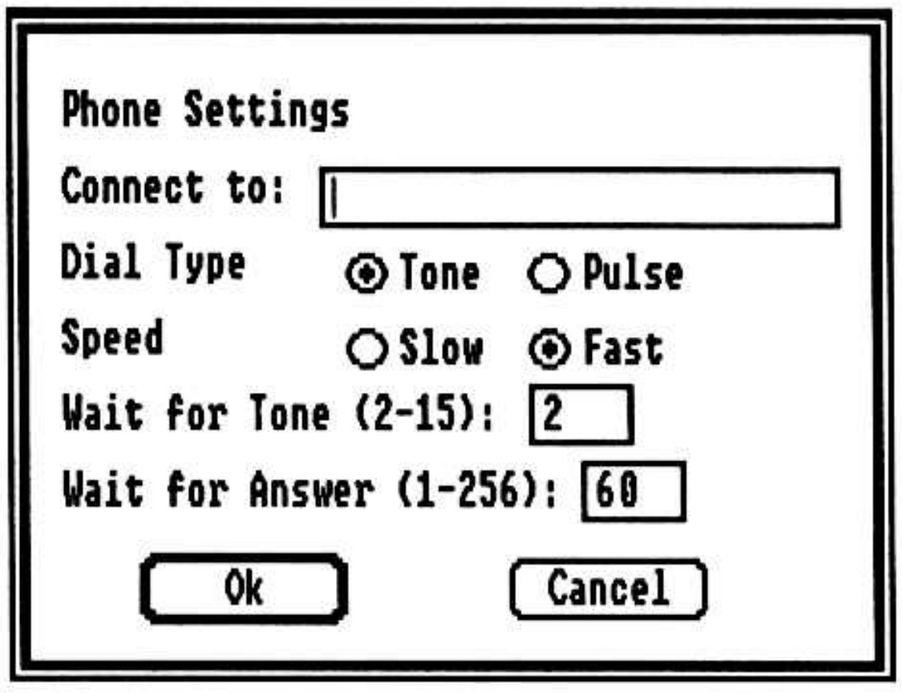


Figure 8.3 Labels on Controls

Chapter 9
Message Boxes

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9.1 Introduction

A message box is a special dialog box through which an application can display error messages and other important information. Message boxes are modal dialog boxes and require immediate attention by the user before work with the application can proceed. Applications developed for the Microsoft Windows operating environment should use message boxes only when an error occurs or when the application must supply information that the user needs to complete an action.

9.2 Message-Box Types

There are four types of message boxes. Each message box has an appropriate icon that should be located in the upper left corner of the dialog box.

Icon	Message Box	Description
	Status	Informs the user that a task is in progress. It also can specify an action that the user must complete before a task can proceed. This message box contains no choices for the user to make.
	Note	Reminds the user of the action about to be taken. This message box asks for a user response and may contain choices the user can act on.
	Warning or Caution	Warns the user of consequences of carrying out an action and requires the user to give instruction on how to proceed. This message box should be used only when the user must be warned of something. The box often contains Yes, No, and Cancel buttons. This message box

		usually appears when data could be lost as a result of proceeding, for example, when the user deletes a file.
	Stop	Informs the user of a serious system-related deficiency that must be remedied before any action can be carried out.

Only the Stop message box should be system-modal. All other message boxes should be application-modal.

Note

Applications should use system-modal message boxes only when they encounter a condition that is the result of a system-related problem or when the condition may dramatically affect other applications running on the system.

9.3 The System Beep

Applications may use the system beep to signal a minor and obvious error, such as moving from the client area into a menu with the mouse button pressed.

9.4 Message-Box Captions

At a minimum, every message box should have a title bar and caption. The caption should include only the name of the application. For example, applications should not use "WARNING" or exclamation points in a caption. The text of the message box should clearly specify whether or not the message is a warning or error message.

If the message box indicates a system-related error, the caption should be as follows:

```
System Error
```

9.5 Push Buttons in Message Boxes

Message boxes should contain only push buttons. Other control types are inappropriate in message boxes. Windows provides the following sets of push buttons for use with message boxes:

Push Button	Description
Ok	Use when no explicit action is required.
Ok or Cancel	Use when Ok is the default and the user has the option of canceling the command.
Yes or No	Use when the default is No and the user has the option of proceeding with the command.
Yes, No, or Cancel	Use when the user has the Yes and No options within the command, as well as the option to cancel the command and proceed with the application.
Retry or Cancel	Use when a device error occurs, such as no disk in the disk drive. The user should cancel the command or retry the operation that causes the error. Applications must not let the user ignore any device error.

The best or safest response should always be shown as the default button. If Ok is used, it should be the default button.

Buttons should never be ambiguous or redundant. For example, if an error message requires only acknowledgment, the Ok button is sufficient; a Cancel button is unnecessary.

9.6 Keyboard Use

Message boxes use the same keyboard support as other dialog boxes. The TAB key moves between buttons, the SPACEBAR activates the selected button, the ENTER key activates the default button, and the ESCAPE key cancels the message box (when the box contains a Cancel button).

9.7 Flashing for Attention

Any application that wishes to display a message box when it or any of its child windows is not currently active should flash its title bar or icon to attract the attention of the user. The application should postpone displaying the message box until the user makes the application active.

On system-critical or potentially unrecoverable errors, the application should display a message box. No flashing is recommended.

9.8 Sample Message Boxes

The following are some sample message boxes that can be shared by applications that have standard File menus:

- If the user selects the New or Open... command in the File menu after changes have been made during the current session, a message box with the following format should be displayed:

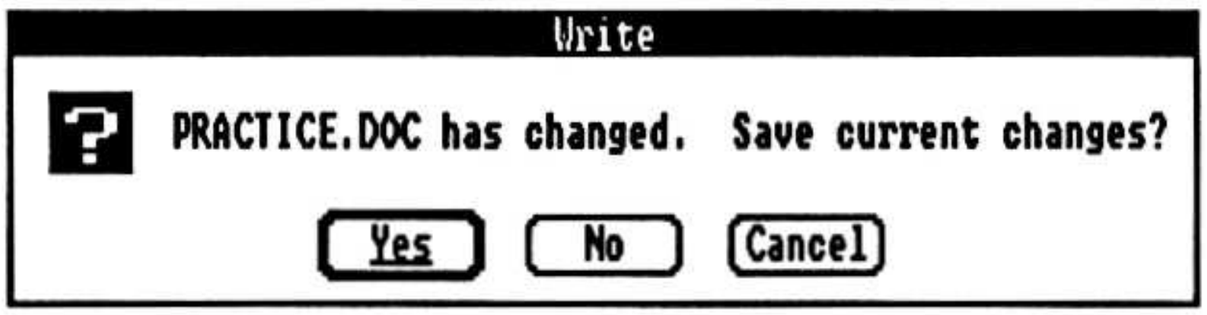


Figure 9.1 Dialog Box with Warning

- If the user selects the Save As... command and then attempts to save the file under the name of an existing file, a message box with the following format should be displayed:

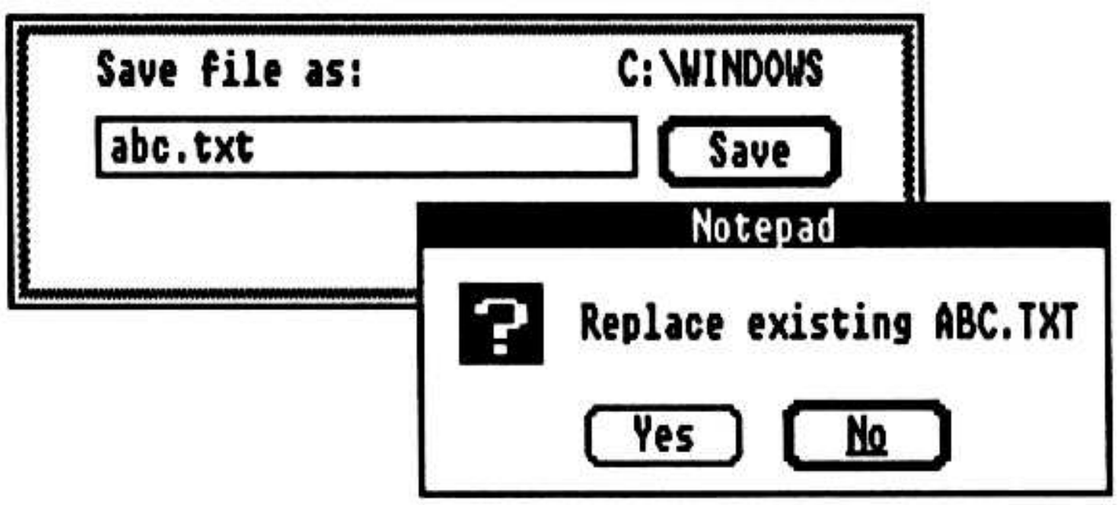


Figure 9.2 Save As... Dialog Box with Warning

- If the user invokes an application with the name of a data file that does not exist, a message box with the following format should be displayed:



Figure 9.3 Dialog Box with Warning

If the user selects Ok, the default button, the application should display the filename in the title bar and as the default field in a Save As... dialog box. If the user selects Cancel, the application should ignore the filename and continue operation as if no file had been specified.

If the user selects Ok, and the application displays the filename, it should also attach the standard extension unless the user supplies one.

- If an application has a Print command, it should display a print message box whenever it sends a file to the spooler for printing. The caption should contain the application's name only. The dialog box should appear in the following format:



Figure 9.4 Dialog Box with Print Message

The name of the file appears in uppercase letters. The dialog box should also have a centered Cancel button.

9.9 Style Suggestions for Message-Box Text

This section contains suggestions for text format that you may want to use in creating message boxes. Following these guidelines will help ensure a consistent user interface.

Message-box text should describe the problem or error as clearly as possible. The message may suggest how to remedy the problem, but it should not be longer than 2-3 lines of text. Technical descriptions of the problem should not be used.

When writing text for a message box, do the following:

- Use "Cannot" instead of "Can not" or "Can't."
- Use "Not enough ... to ...." instead of "Disk is full, save not completed." For example:

  ```
  Not enough disk space to save PICTURE.MSP.
  ```

- Always use an ending period or question mark.

- Do not put the name of the application in the message.
- Do not imbed colons in the message.
  The following example is correct:
  `Cannot read X.DAT.`
  This example is incorrect:
  `Cannot read: X.DAT.`
- Use "Filename" instead of "File name."

Chapter 10
Cooperation

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10.1 Introduction

Cooperation between applications is the key element in making the Microsoft Windows operating environment a shared environment for applications. Applications should do the following:

- Use the CPU sparingly.

  Applications that perform extended computations should yield control whenever possible to let other applications complete their tasks.

- Use the screen display only when active.

  Inactive applications should alert the user with a flashing title bar or icon if they need to display a dialog box. The user then can activate the application.

- Use only as much memory as needed.

- Use discardable code segments.

  Applications that consist of small, discardable code segments are much easier to move and swap when available memory is short.

- Avoid fixed memory objects.

10.2 Loading and Initialization

Applications should display their initial screen setup as soon as possible after loading. They should not leave a window empty for a long period of time.

10.3 Yielding and Time-Critical Operations

All applications should yield control whenever possible to allow other applications to access the CPU. For reasonably short operations where control is crucial, such as loading or saving a file, applications should complete the full operation as quickly as possible.

10.4 System Hourglass

All applications should display the system-hourglass pointer when they begin lengthy or dead-air (pause in activity) operations. This informs the user that interaction with the application is not currently possible. The hourglass should be displayed with both keyboard and mouse interfaces. The hourglass pointer is shown as follows:



10.5 Dead Air

Unless waiting for user input, applications with the input focus must not present any pause in activity (dead air) that lasts longer than five seconds. Applications that require long delays in some operations should signal the user in some visible way that the operation is progressing.

10.6 Memory Usage

Applications should cooperate in using memory. Applications should always determine whether there is sufficient space to operate in RAM as well as on the disk. If there is not, the application should display an appropriate message to let the user know that it is not possible to run in the current environment.

Applications should be designed to use only as much memory as is reasonably required to run the application.

10.7 File Usage

Applications should always close files located on a floppy disk when not explicitly carrying out disk operations. Leaving these files open can cause a system failure. It is an application's responsibility to ensure that errors, such as the user's removing a disk from the drive without closing a file, do not cause the system to fail.

10.8 Color

Applications should use color wherever appropriate; however, consideration also must be given to how the program looks on monochrome displays.

Chapter 11
International Concerns

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11.1 Introduction

Software companies are reaching out to international markets for increased sales. Competition from local products has increased dramatically. To reach these markets, applications developed for the Microsoft Windows operating environment should be fully localized. You should design your product from the start with international markets in mind.

This chapter summarizes the localization concerns you should consider when writing an application. Following these instructions will make your application easy to translate into various languages.

11.2 Resource Files

If the following are used in an application, they should be placed in the application's resource file: menu items, menu bars, dialog boxes, cursors, icons, bitmaps, fonts, strings, and accelerators. This will allow a non-technical person to translate menus and messages and to resize and change the layout of dialog boxes. This person does not have to know anything about the source code of your application. Including this information in the resource file will dramatically cut the time required to bring your application to a new market.

11.3 Character Sets and Keyboards

Foreign countries use characters that are above 127 in the ANSI character set. These characters do not appear on US keyboards, and keyboard layouts may be very different from the US keyboard. These differences can lead to difficulty when selling products in international markets.

Since Windows uses the ANSI character set, you should consider adding a conversion utility to your application. In international markets, you will always be using characters above 127 from the 8-bit character set. Below 127, there is no difference between ASCII and ANSI. Above 127, you will encounter problems if you are loading a file created using the ASCII set into your Windows application. You should use the **OemToAnsi** and **AnsiToOem** functions to create a two-way conversion utility. The

conversion utility also should recognize any file that it did not create. It should then invoke a dialog box to ask if the user wants to convert that file into the application's Windows format.

11.4 Formats

Different countries use different formats for numbers, currencies, time, dates, and measures, so porting an application to another country may be difficult. Windows provides you with a way to solve this problem. When requesting any of the above-mentioned values, your application should go to the *win.ini* file. *Win.ini* has an international section that contains all the values for the particular country Windows is installed for. At the time of installation, the **setup** program creates an international table in *win.ini* and puts all these values in it. The values entered are based on the user's choice of a particular country's keyboard. The following is a list of the values set in the [intl] section in *win.ini*:

Settings	Description
`iCountry=`	Country Code, see your DOS manual for details.
`iDate=`	0 for mdy, 1 for dmy, 2 for ymd.
`iCurrency=`	0 for currency symbol prefix, no separation. 1 for currency symbol suffix, no separation. 2 for currency symbol prefix, one-character separation. 3 for currency symbol suffix, one-character separation.
`iDigits=`	Number of significant decimal digits in currency.
`iTime=`	0 for 12 hour clock, 1 for 24 hour clock.
`iLzero=`	0 for no leading zeros, 1 for leading zeros.
`s1159=`	Trailing string from 0:00 to 11:59.
`s2359=`	Trailing string from 12:00 to 23:59.
`sCurrency=`	Currency-symbol string.
`sThousand=`	Thousands-separator string.
`sDecimal=`	Decimal-separator string.

`sDate=`	Date-separator string.
`sTime=`	Time-separator string.
`sList=`	List-separator string.
`dialog=`	Is always "yes." This will activate the Country Settings Dialog Box.

These format settings must not be hardcoded. An application must get these values from the *win.ini* file.

The following is a sample file for the United States from the "[intl]" section in *win.ini*:

```
[intl]
iCountry=1
iDate=0
iCurrency=0
iDigits=2
iTime=0
iLzero=0
s1159=AM
s2359=PM
sCurrency=$
sThousand=,
sDecimal=.
sDate=-
sTime=:
sList=,
dialog=yes
```

11.5 Additional Software Design Tips

- An application should include the following in its resource file:

  Any text string that the user may see in the form of a screen message.

  All dialog-box text and coordinates.

  All child-window text and coordinates.

- The application's dialog boxes and child windows should not take up more than 75% of the screen space. This restriction allows the enlargement of the dialog box and/or child window when translating the message strings.

- An application should never rely on the position of a string in a menu, dialog box, or child window. This saves you from having to modify the source code if the position of the string changes when it is translated.

- An application should never rely on the length of a string in a menu, dialog box, or child window. When translated, the length of the string may change.

- If a string contains variables, these variables should be relocatable anywhere within the body of the string. In different languages, the position of a word in a string may vary. The recommended way to represent variables is to use "%%" as a placeholder. The "%%" placeholder then can be relocated by the translator anywhere within the string. An example of this is the following system message:

  ```
  COPYING FOO.DOC
  ```

  The variable is *foo.doc*. In French, this becomes:

  ```
  COPIE DE FOO.DOC EN COURS
  ```

  The variable *foo.doc* had to be relocated in the middle of this string. In the *msdos.rc* file, the string appears as:

  ```
  "COPYING %%"
  ```

- Accelerator keys should not be hardcoded. They should be modifiable in the application's resource file.

- ALT–*key* combinations should be examined carefully. Some international keyboards use certain ALT–*key* combinations to enter certain characters.

- Application names, as well as filenames and extensions produced by the application, should not be hardcoded. They should be modifiable in the application's resource file.

- Applications should not use any OEM-specific virtual keys, with the exception of alphanumeric keys. Using these OEM-specific virtual keys prevents you from porting the application to international markets that use different keyboard layouts, unless you make substantial code modifications.

- Applications should accept and output the full range of the 8-bit ANSI character set. Your application should not be dependent on the US character set or any specific country's character set.

- An application should support both metric and English measurement systems.
- If an application is doing any kind of sorting, it should account for the various international sorting schemes.

Chapter 12
Help Facility

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12.1 Introduction

To allow users to request on-line information about your application, you may decide to include a help facility in the application. If on-line help is available, the user must be able to access it both from the keyboard and by using the mouse (or similar pointing device).

There are two main types of help available: general information and context-sensitive information.

12.2 General Help

The application should allow the user to obtain general help information by choosing the Help command from the System menu of the application. When the Help command is chosen, the application should display a dialog box with the following form:

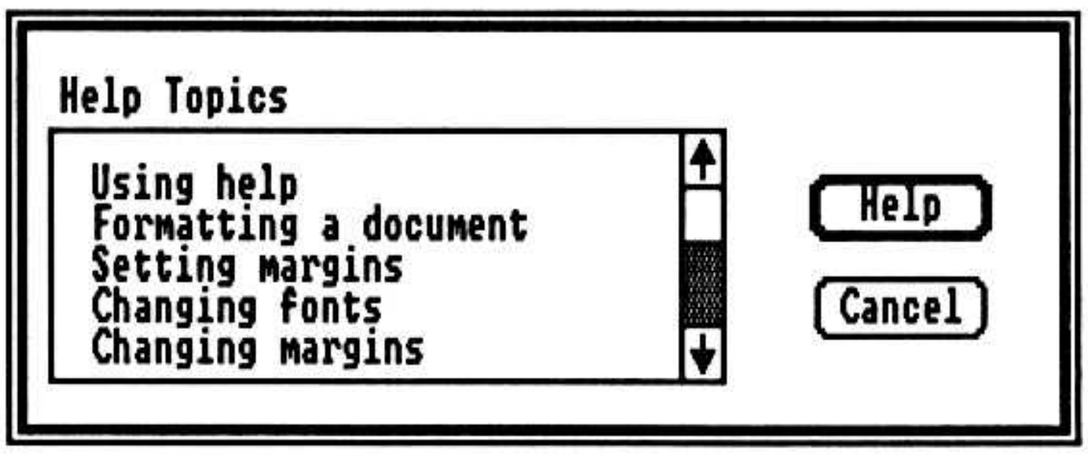


Figure 12.1 Help Topics Dialog Box

The dialog box contains a list box showing all of the topics the user can get information about. The user should be able to choose a topic by any of the following methods:

- Choose the Help push button or press ENTER to choose the currently highlighted topic.
- Double-click a topic with a mouse or other pointing device.
- Select a topic and then choose the Help button or press ENTER.

The Help push button should be the default button. If the list box

contains more topics than can be shown in the list box, the dialog box should have scroll bars to allow the user to request additional topics. If the user presses the Cancel button, the application should dismiss the Help dialog box and resume the application.

When the user chooses a topic by any of the previous methods, the application should display a dialog box that can be sized and moved. This dialog box contains the help information about the selected topic. The dialog box has the following form:

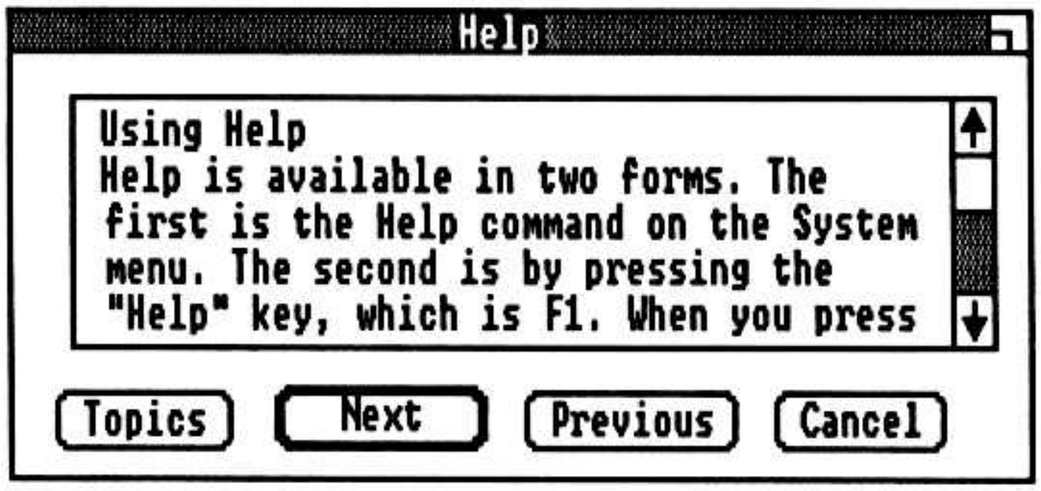


Figure 12.2 Help Information Dialog Box

If the user chooses the Topics button, the application should display the Help Topics dialog box. If the user chooses Next, the application should display the Help Information dialog box for the next topic on the list. The Next button is the default button. If the user chooses the Previous button, the application should display the Help Information dialog box for the previous topic. Choosing the Cancel button or pressing the ESCAPE key should return the user to the previous location in the application.

If more information is available than can be displayed in the list box, scroll bars should be provided to allow the user to view the rest of the text.

12.3 Context-Sensitive Help

Another way the user can request help while using an application is to press the F1 key. This places the application in "help mode." In this mode, the next item the user selects (menu, scroll bar, control, etc.) should display a Help Information dialog box that gives information about the selected item. This is known as context-sensitive help. The dialog box has

the same format as the one shown in the preceding Figure 12.2.

When the user presses the F1 key, the mouse pointer changes to a question mark. If no mouse is installed, the question-mark pointer is still displayed in the middle of the application window. If the application is displaying a dialog box when the F1 key is pressed, a Help Information dialog box that lists information about the application's dialog box should be displayed automatically.

When the user presses the ESCAPE key, the application should cancel the help mode and display the user's previous location in the application.

12.4 Prompts in Dialog Boxes

In addition to the two main types of help information, there is one other you may want to add to your application. In some of your application's dialog boxes, you may want to display a suggestion prompt such as the following:

```
For help, press F1
```

Generally, this prompt line should be placed at the bottom of the dialog box to keep the prompt line close to the command area (where Ok, Cancel, and other buttons are located). A prompt line in a dialog box should not be highlighted. To avoid screen clutter, these prompt lines should be used sparingly.

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